

The topic is

- The topic is a general term for the subject or theme of a text, speech, or conversation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a statement.
- The topic can be identified by looking for clues in the text, such as the title, the introduction, the main idea, the keywords, or the summary.
- The topic can be used to organize and structure the information in the text, such as by using headings, subheadings, paragraphs, or bullet points.
- The topic can be used to guide the research and analysis of the text, such as by finding relevant sources, evaluating the evidence, and drawing conclusions.
- The topic can be used to communicate the purpose and message of the text, such as by using a thesis statement, a hook, a transition, or a call to action.

FUNDAMENTALS OF MECHANICAL ENGINEERING

- Mechanical engineering is the study of physical machines that may involve force and movement.
- It is an engineering branch that combines engineering physics and mathematics principles with materials science, to design, analyze, manufacture, and maintain mechanical systems.
- It is one of the oldest and broadest of the engineering branches.
- Mechanical engineering, as its name suggests, deals with the mechanics of operation of mechanical systems.
- The mechanical engineer may design a component, a machine, a system or a process.
- Mechanical engineers will analyze their design using the principles of motion, energy, and force to ensure the product functions safely, efficiently, reliably, and can be manufactured at a competitive cost.
- Mechanical engineering plays a dominant role in enhancing safety, economic vitality, enjoyment and overall quality of life throughout the world.
- Mechanical engineers are concerned with the principles of force, energy and motion.
- The Fundamentals of Engineering (FE) exam is generally the first step in the process to becoming a professional licensed engineer (P.E.).
- It is designed for recent graduates and students who are close to finishing an undergraduate engineering degree from an EAC/ABET-accredited program.

Unit 1 - Introduction to Mechanics

- Mechanics is the branch of physics that deals with the motion of objects and the forces that act on them.
- Mechanics can be divided into two main subfields: kinematics and dynamics.
- Kinematics is the study of how objects move without considering the causes of their motion.
- Dynamics is the study of how forces affect the motion of objects.
- Mechanics can also be classified into two types: classical and modern.
- Classical mechanics is based on the laws of motion formulated by Isaac Newton in the 17th century. It describes the motion of objects that are relatively large and slow compared to the speed of light.
- Modern mechanics is based on the theories of relativity and quantum mechanics developed in the 20th century. It describes the motion of objects that are very small or very fast compared to the speed of light.
- Some of the applications of mechanics include engineering, astronomy, sports, biomechanics, robotics, etc.

Force, Moment and Couple

Force

- A force is a push or pull that acts on a body and changes or tends to change its state of rest or motion.
- A force is a vector quantity, meaning it has both magnitude and direction.
- A force can be represented by an arrow whose length is proportional to the magnitude and whose direction is the same as the direction of the force.
- A force can be resolved into components along any two perpendicular axes, such as x and y, using the trigonometric functions of the angle between the force and the axes.
- The net force acting on a body is the vector sum of all the forces acting on it. If the net force is zero, the body is in equilibrium.

Moment

- A moment is the tendency of a force to rotate a body about a point or an axis.
- A moment is also a vector quantity, meaning it has both magnitude and direction.
- A moment is also called a torque or a couple.
- The magnitude of a moment is equal to the product of the force and the perpendicular distance from the point or axis of rotation to the line of action of the force.
- The direction of a moment is perpendicular to the plane containing the force and the point or axis of rotation, and follows the right-hand rule:

if the fingers of the right hand curl in the direction of the rotation, the thumb points in the direction of the moment.

- The moment of a force is only defined with respect to a certain point or axis of rotation, and in general, it changes when the point or axis of rotation changes.
- The net moment acting on a body about a point or an axis is the vector sum of all the moments acting on it. If the net moment is zero, the body is in rotational equilibrium.

Couple

- A couple is a system of two equal and opposite parallel forces that act on a body and produce a moment but no net force.
- A couple is also a vector quantity, meaning it has both magnitude and direction.
- The magnitude of a couple is equal to the product of one of the forces and the perpendicular distance between the lines of action of the two forces. This distance is also called the arm of the couple.
- The direction of a couple is perpendicular to the plane containing the two forces, and follows the right-hand rule: if the fingers of the right hand curl in the direction of the rotation, the thumb points in the direction of the couple.
- The moment of a couple is independent of the point or axis of rotation, meaning it is the same for any point or axis of rotation.
- A couple can be resolved into a force and a moment at any point by adding a force equal and opposite to one of the forces of the couple at that point. This process is called resolving a force into a force and a couple.
- A couple can also be replaced by another couple with the same magnitude and direction but with a different arm. This process is called shifting a couple.

Principle of Transmissibility

- The principle of transmissibility states that the point of application of a force can be moved anywhere along its line of action without changing the external reaction forces on a rigid body .
- The line of action of a force is the infinite straight line that passes through the point of application of the force and is parallel to the direction of the force.
- The principle of transmissibility is based on the assumption that the rigid body is in equilibrium under the action of the forces, and that the internal stresses and strains are negligible.
- The principle of transmissibility can be used to simplify the analysis of concurrent force systems, where all the forces intersect at a common point.
- The principle of transmissibility can also be used to replace a force by

an equivalent force-couple system, where the force is moved to a different point along its line of action and a couple is added to balance the moment caused by the displacement of the force.

- The principle of transmissibility has some limitations, such as:
 - It does not apply to non-rigid bodies, where the deformation and internal forces depend on the point of application of the force.
 - It does not apply to frictional forces, where the point of application of the force affects the normal reaction and the friction coefficient.
 - It does not apply to distributed forces, where the force is not concentrated at a single point but acts over a region of the body.

Varignon's Theorem

- Varignon's theorem is a useful tool in scalar moment calculations. It states that the moment of a resultant of two concurrent forces about any point is equal to the algebraic sum of the moments of its components about the same point .
- Varignon's theorem can be extended to any number of concurrent forces. It can also be applied to couples, since a couple can be considered as two equal and opposite forces.
- Varignon's theorem can be used to simplify the calculation of moments when the perpendicular distance is hard to determine or when the forces are not in the same plane.
- Varignon's theorem can be derived from the definition of moment and the principle of superposition. The proof is based on the fact that the moment of a force about a point is equal to the cross product of the position vector of the point of application of the force and the force vector .
- Varignon's theorem can also be interpreted as a statement in Euclidean geometry, that deals with the construction of a particular parallelogram, the Varignon parallelogram, from an arbitrary quadrilateral. It is named after Pierre Varignon, whose proof was published posthumously in 1731.
- Varignon's theorem can be illustrated by the following diagram:

Varignon's theorem diagram

- In the diagram, A, B, C, and D are four arbitrary points in a plane. E, F, G, and H are the midpoints of the sides AB, BC, CD, and DA, respectively. Varignon's theorem states that the quadrilateral EFGH is a parallelogram, and that its area is half the area of the quadrilateral ABCD.

Resultant of Force System- Concurrent and Non-concurrent Coplanar Forces

- A force system is a set of forces acting on a rigid body.

- A resultant force is a single force that has the same effect as the original force system.
- A coplanar force system is a force system in which all the forces lie in the same plane.
- A concurrent force system is a force system in which all the lines of action of the forces intersect at a common point.
- A non-concurrent force system is a force system in which the lines of action of the forces do not intersect at a common point.
- A parallel force system is a force system in which all the forces are parallel to each other.
- A non-parallel force system is a force system in which the forces are not parallel to each other.

Resultant of Concurrent Coplanar Force System

- The resultant of a concurrent coplanar force system can be found by using the principle of vector addition.
- The principle of vector addition states that the resultant of two or more vectors is the vector obtained by placing the vectors head to tail and drawing a vector from the tail of the first vector to the head of the last vector.
- The resultant of a concurrent coplanar force system can also be found by using the method of rectangular components.
- The method of rectangular components states that any vector can be resolved into two perpendicular components, such as horizontal and vertical components, and the resultant of the vector is the vector sum of its components.
- The resultant of a concurrent coplanar force system can be expressed in terms of its magnitude and direction.
- The magnitude of the resultant can be found by using the Pythagorean theorem: $R = \sqrt{R_x^2 + R_y^2}$, where R_x and R_y are the horizontal and vertical components of the resultant, respectively.
- The direction of the resultant can be found by using the trigonometric function: $\tan \theta = R_y / R_x$, where θ is the angle between the resultant and the horizontal axis.

Resultant of Non-concurrent Coplanar Force System

- The resultant of a non-concurrent coplanar force system can be found by using the principle of moments.
- The principle of moments states that the resultant moment of a force system about any point is equal to the algebraic sum of the moments of the individual forces about the same point.
- The moment of a force about a point is the product of the magnitude of the force and the perpendicular distance from the line of action of the force to the point.

- The resultant of a non-concurrent coplanar force system can be expressed in terms of its resultant force and resultant moment.
- The resultant force can be found by using the method of rectangular components, as explained above.
- The resultant moment can be found by using the formula: $M = \Sigma(F \times d)$, where F is the magnitude of the force, d is the perpendicular distance from the line of action of the force to the point of interest, and the summation is taken over all the forces in the system.
- The point of interest can be any point in the plane, but it is usually chosen to be the point where the resultant force intersects the plane, which is called the point of concurrency.
- The point of concurrency can be found by using the formula: $x = -M_y / R$ and $y = M_x / R$, where x and y are the coordinates of the point of concurrency, M_y and M_x are the moments of the resultant force about the x -axis and y -axis, respectively, and R is the magnitude of the resultant force.

Types of supports (Hinge, Roller) and loads (Point, UDL, UVL) for the notes of the Unit 1 - Introduction to Mechanics in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

Supports

- Supports are devices that connect a structural member to the ground or to another part of the structure and provide resistance to forces and moments.
- Supports can be classified into four types: simple, roller, hinge, and fixed.
- Simple support: A simple support allows rotation about any axis and translation (horizontal movement) in any direction, but prevents vertical movement. It can resist only a vertical force. An example of a simple support is a cable or a rope.
- Roller support: A roller support allows rotation about any axis and translation in any direction parallel to the surface on which it rests, but prevents translation perpendicular to the surface. It can resist only a vertical force. An example of a roller support is a wheel or a cylinder.
- Hinge support: A hinge support allows rotation about any axis, but prevents translation in any direction. It can resist both vertical and horizontal forces, but not moments. An example of a hinge support is a pin or a bolt.
- Fixed support: A fixed support prevents rotation and translation in any direction. It can resist both vertical and horizontal forces and moments. An example of a fixed support is a wall or a foundation.

Loads

- Loads are external forces or moments that act on a structural member and cause deformation, stress, or displacement.
- Loads can be classified into three types: point, uniformly distributed (UDL), and uniformly varying (UVL).
- Point load: A point load is a load that acts at a single point on a structural member. It can be represented by a single force or moment vector. An example of a point load is a weight or a concentrated force.
- Uniformly distributed load (UDL): A uniformly distributed load is a load that acts with a constant intensity over a length or area of a structural member. It can be represented by a series of parallel force vectors or a single equivalent force vector. An example of a UDL is a self-weight or a uniform pressure.
- Uniformly varying load (UVL): A uniformly varying load is a load that acts with a varying intensity over a length or area of a structural member. It can be represented by a series of non-parallel force vectors or a single equivalent force vector. An example of a UVL is a triangular load or a linearly varying pressure.

Free Body Diagram

A free body diagram is a graphical tool that helps us to analyze the forces acting on an object in a given situation. It is a simplified representation of the object and the forces, without showing the details of the external environment or the internal structure of the object. A free body diagram can help us to apply Newton's laws of motion and solve problems involving equilibrium, motion, and statics.

To draw a free body diagram, we need to follow these steps:

- Identify the object of interest and isolate it from the surroundings. Draw a simple shape to represent the object, such as a dot, a box, or a circle.
- Identify all the forces acting on the object and label them with arrows pointing in the direction of the force. The length of the arrow should indicate the relative magnitude of the force. The label should indicate the type and source of the force, such as gravity, tension, friction, normal, etc.
- If the object is in contact with another object or a surface, include the reaction forces that the object experiences due to the contact. For example, if the object is resting on a table, there is a normal force exerted by the table on the object, and a friction force if there is relative motion between the object and the table.
- If the object is connected to another object by a string, rope, or cable, include the tension force that the string exerts on the object. The tension force is always directed along the string and away from the object.
- If the object is subjected to a distributed force, such as air resistance or

fluid pressure, replace it with an equivalent point force that has the same net effect on the object.

- If the object is rotating or has angular acceleration, include the torque or moment that causes the rotation. The torque is a vector quantity that has a direction perpendicular to the plane of rotation and follows the right-hand rule.

Here is an example of a free body diagram for a block sliding down an inclined plane:

free body diagram example

The forces acting on the block are:

- The weight of the block, which is a gravitational force directed vertically downward and equal to the mass of the block times the acceleration due to gravity.
- The normal force of the plane, which is a contact force directed perpendicular to the plane and equal to the component of the weight that is perpendicular to the plane.
- The friction force of the plane, which is a contact force directed parallel to the plane and opposite to the direction of motion. It is proportional to the normal force and the coefficient of friction between the block and the plane.
- The applied force, which is an external force that pushes or pulls the block along the plane. It can have any direction and magnitude, depending on the situation.

The free body diagram can help us to write the equations of motion for the block, using Newton's second law, which states that the net force on an object is equal to its mass times its acceleration. We can resolve the forces into components along the x-axis (parallel to the plane) and the y-axis (perpendicular to the plane), and write:

$$\sum F_x = F - f - mg \sin \theta = ma_x$$

$$\sum F_y = N - mg \cos \theta = ma_y$$

where F is the applied force, f is the friction force, mg is the weight of the block, θ is the angle of the plane, N is the normal force, m is the mass of the block, a_x is the acceleration along the x-axis, and a_y is the acceleration along the y-axis.

By solving these equations, we can find the values of the unknown forces and the acceleration of the block.

Equilibrium Equations and Support Reactions

- Equilibrium equations are the conditions that must be satisfied for a structure or a body to be in static equilibrium, i.e., to have no net force or moment acting on it.
- The equilibrium equations for a two-dimensional system are:
 - Sum of forces in the x-direction is zero: $\sum F_x = 0$
 - Sum of forces in the y-direction is zero: $\sum F_y = 0$
 - Sum of moments about any point is zero: $\sum M = 0$
- The equilibrium equations for a three-dimensional system are:
 - Sum of forces in the x-direction is zero: $\sum F_x = 0$
 - Sum of forces in the y-direction is zero: $\sum F_y = 0$
 - Sum of forces in the z-direction is zero: $\sum F_z = 0$
 - Sum of moments about the x-axis is zero: $\sum M_x = 0$
 - Sum of moments about the y-axis is zero: $\sum M_y = 0$
 - Sum of moments about the z-axis is zero: $\sum M_z = 0$
- Support reactions are the forces and moments that are applied by the supports or constraints on a structure or a body to keep it in equilibrium.
- The types of support reactions depend on the types of supports or constraints, such as:
 - A pin or a hinge support can provide two reaction forces in the x- and y-directions, but no moment.
 - A roller or a rocker support can provide one reaction force in the direction perpendicular to the surface, but no moment.
 - A fixed or a built-in support can provide two reaction forces in the x- and y-directions, and one reaction moment.
 - A cable or a rope support can provide one reaction force in the direction of the cable or rope, but no moment.
- To compute the support reactions for a planar structure, the following steps can be followed:
 - Draw a free-body diagram of the structure, showing all the applied loads and the support reactions as unknowns.
 - Apply the equilibrium equations to the free-body diagram, and solve for the unknown support reactions.
 - If the number of unknowns is equal to the number of equilibrium equations, the structure is statically determinate, and the support reactions can be found uniquely.
 - If the number of unknowns is more than the number of equilibrium equations, the structure is statically indeterminate, and the support reactions cannot be found by equilibrium equations alone. Additional methods, such as compatibility equations, are needed.

- If the number of unknowns is less than the number of equilibrium equations, the structure is unstable, and the support reactions cannot be found. The structure will collapse or move under any load.

Normal and Shear Stress

- Normal stress is the stress that acts perpendicular to the cross-sectional area of a material. It arises from the normal force, the component of force vector perpendicular to the material cross section. Normal stress can be either tensile or compressive, depending on the direction of the normal force. Tensile stress causes elongation of the material, while compressive stress causes shortening of the material.
- Shear stress is the stress that acts parallel to the cross-sectional area of a material. It arises from the shear force, the component of force vector parallel to the material cross section. Shear stress causes deformation of the material, such as twisting, sliding, or shearing .
- Normal and shear stress are related by the following equation:

$$\tau = \frac{V}{A}$$

where τ is the shear stress, V is the shear force, and A is the cross-sectional area of the material. - Normal and shear stress can be illustrated by the following examples:

- A light fixture hanging from the ceiling by a rope experiences both normal and shear stress. The normal stress is due to the weight of the fixture, which acts perpendicular to the cross-sectional area of the rope. The shear stress is due to the tension in the rope, which acts parallel to the cross-sectional area of the rope.
- A beam supported by two columns experiences both normal and shear stress. The normal stress is due to the bending moment, which causes compression on the upper part of the beam and tension on the lower part of the beam. The shear stress is due to the shear force, which causes deformation of the beam along its length.

Strain

Strain is a measure of the amount of deformation that an object undergoes due to an applied force. Strain is a dimensionless quantity that is calculated by dividing the change in length or shape of an object by its original length or shape. Strain can be positive or negative, depending on whether the object is stretched or compressed.

There are three main types of strain:

- Longitudinal strain: This is the strain that occurs when an object is subjected to a force that is parallel to its length. Longitudinal strain is defined as the ratio of the change in length to the original length of the object. For example, if a metal rod of length L is stretched by a force F and its length becomes $L + \Delta L$, then the longitudinal strain is $\Delta L/L$. Longitudinal strain can be tensile (positive) or compressive (negative).
- Volumetric strain: This is the strain that occurs when an object is subjected to a force that changes its volume. Volumetric strain is defined as the ratio of the change in volume to the original volume of the object. For example, if a balloon of volume V is inflated by a pressure P and its volume becomes $V + \Delta V$, then the volumetric strain is $\Delta V/V$. Volumetric strain can be positive (expansion) or negative (contraction).
- Shearing strain: This is the strain that occurs when an object is subjected to a force that causes it to slide or twist. Shearing strain is defined as the ratio of the change in angle to the original angle of the object. For example, if a book of thickness t is pushed by a force F and its top surface slides by a distance x , then the shearing strain is x/t . Shearing strain is always positive.

Hookes' law

- Hookes' law is a law of elasticity that relates the force applied to a spring or other elastic object to the amount of deformation or displacement it undergoes.
- Hookes' law can be expressed mathematically as $F = kx$, where F is the force, k is the spring constant or stiffness, and x is the displacement or extension of the spring from its equilibrium position.
- Hookes' law is valid only for small deformations, where the force and displacement are proportional. For larger deformations, the relationship becomes nonlinear and more complex.
- Hookes' law can be used to model the behavior of many elastic materials, such as metals, rubber, wood, etc., as well as natural phenomena, such as the oscillation of a pendulum, the vibration of a guitar string, the stretching of a DNA molecule, etc.
- Hookes' law was discovered by the English scientist Robert Hooke in 1660, who observed that the extension of a spring is proportional to the tension applied to it. He published his findings in his book *De Potentia Restitutiva*, or *Of Spring*, where he also coined the term *elasticity*.
- Hookes' law is one of the fundamental principles of mechanics, and it forms the basis of many applications in engineering, physics, biology, and other fields.

Poisson's Ratio

- Poisson's ratio is a measure of the Poisson effect, the deformation (expansion or contraction) of a material in directions perpendicular to the specific direction of loading.
- Poisson's ratio is defined as the negative ratio of the transverse strain to the longitudinal strain of a material when it is stretched or compressed .
- Mathematically, Poisson's ratio can be represented as:

$$\mu = -\frac{\epsilon_{lat}}{\epsilon_{long}}$$

Where,

μ = Poisson's ratio

ϵ_{lat} = Lateral strain

ϵ_{long} = Longitudinal strain

- Lateral strain is the change in the dimension of a material perpendicular to the applied force, divided by the original dimension.
- Longitudinal strain is the change in the dimension of a material parallel to the applied force, divided by the original dimension.
- Poisson's ratio is a dimensionless quantity and has no unit.
- Poisson's ratio can have values between -1 and 0.5 for most materials.
- Poisson's ratio is positive for materials that contract in the transverse direction when stretched in the longitudinal direction, and negative for materials that expand in the transverse direction when stretched in the longitudinal direction.
- Poisson's ratio is an important parameter for describing the mechanical behavior of materials and designing structures and devices that involve deformation and stress.
- Some examples of Poisson's ratio for different materials are:

Material	Poisson's ratio
Rubber	0.5
Steel	0.3
Cork	0
Corkscrew	-0.7

Elastic Constants and Their Relationship

Elastic constants are the ratios of the applied stress to the resulting strain in an elastic body. They describe the elastic behavior of objects under different types of deformation, such as tension, compression, shear, or torsion. There are four

main elastic constants: Young's modulus (E), bulk modulus (K), shear modulus (G), and Poisson's ratio (ν).

- Young's modulus (E) is the ratio of the tensile stress to the tensile strain in a uniaxial deformation. It measures the stiffness of a material in resisting elongation or contraction.
- Bulk modulus (K) is the ratio of the hydrostatic stress to the volumetric strain in a uniform compression or expansion. It measures the resistance of a material to changes in volume.
- Shear modulus (G) is the ratio of the shear stress to the shear strain in a simple shear deformation. It measures the rigidity of a material in resisting shape changes.
- Poisson's ratio (ν) is the ratio of the lateral strain to the longitudinal strain in a uniaxial deformation. It measures the tendency of a material to contract or expand in the direction perpendicular to the applied stress.

These elastic constants are related to each other by the following equations :

- $E = 3K(1 + 2\nu)$
- $E = 9KG / (3K + G)$
- $\nu = (3K - 2G) / (6K + 2G)$

For an isotropic material, which has the same properties in all directions, Poisson's ratio is 0.25 and the elastic constants are related by:

- $E = 3K$
- $E = 2G$

These elastic constants are useful for understanding the mechanical behavior of different materials and designing structures and machines that can withstand various loads and stresses.

Stress-Strain Diagram for Ductile and Brittle Materials

- A stress-strain diagram is a graphical representation of the relationship between the applied stress and the resulting strain in a material under loading.
- The stress is plotted on the vertical axis and the strain is plotted on the horizontal axis.
- The shape and features of the stress-strain diagram can reveal important information about the mechanical properties and behavior of the material.
- Ductile materials are those that can undergo large plastic deformation before fracture, such as metals, polymers and some ceramics.
- Brittle materials are those that fracture with little or no plastic deformation, such as glass, concrete and some ceramics.

- The stress-strain diagrams of ductile and brittle materials have different characteristics, as shown below.

Stress-Strain Diagram for Ductile and Brittle Materials

- The stress-strain diagram of a ductile material typically has the following regions:
 - Elastic region: The initial linear portion of the curve, where the material obeys Hooke's law and the stress is proportional to the strain. The slope of this region is the elastic modulus or Young's modulus of the material.
 - Proportional limit: The point where the linear relationship between stress and strain ends and the material starts to deviate from Hooke's law.
 - Yield point: The point where the material begins to exhibit permanent deformation or plasticity. The stress at this point is the yield strength of the material.
 - Plastic region: The curved portion of the curve, where the material undergoes plastic deformation and the stress increases with increasing strain. The material becomes harder and stronger due to work hardening or strain hardening.
 - Ultimate point: The point where the material reaches its maximum stress or ultimate strength. Beyond this point, the material starts to neck or thin down locally due to the Poisson effect.
 - Fracture point: The point where the material breaks or fractures. The stress at this point is the fracture strength or tensile strength of the material.
- The stress-strain diagram of a brittle material typically has the following regions:
 - Elastic region: The initial linear portion of the curve, where the material obeys Hooke's law and the stress is proportional to the strain. The slope of this region is the elastic modulus or Young's modulus of the material.
 - Proportional limit: The point where the linear relationship between stress and strain ends and the material starts to deviate from Hooke's law.
 - Fracture point: The point where the material breaks or fractures. The stress at this point is the fracture strength or tensile strength of the material.
- The main differences between the stress-strain diagrams of ductile and brittle materials are:
 - Ductile materials have a large plastic region, while brittle materials have little or no plastic region.
 - Ductile materials have a higher ultimate strength than fracture strength, while brittle materials have a lower ultimate strength than fracture strength.
 - Ductile materials show significant necking before fracture, while brittle materials do not.

- Ductile materials show no necking before fracture.
- Ductile materials have a smooth and gradual fracture surface, while brittle materials have a rough and jagged fracture surface.

Factor of Safety

- Factor of safety (FoS) is a measure of how much stronger a system is than it needs to be for an intended load.
- It is the ratio of the ultimate strength (or structural capacity) of a material or a member to the actual working stress or the maximum permissible stress when in use .
- It is a dimensionless quantity that indicates the margin of safety or the degree of reliability of a design.
- It is also known as safety factor (SF) and is sometimes expressed as a percentage.
- It is calculated using detailed analysis, testing, or empirical formulas based on the type of material, load, and application .
- It is influenced by various factors such as material properties, environmental conditions, manufacturing defects, human errors, and design uncertainties .
- It is usually chosen based on codes, standards, regulations, or engineering judgment depending on the level of risk and consequence of failure .
- It is typically greater than one, meaning that the system can withstand more load than the intended one without failure.
- It can also be less than one, meaning that the system is intentionally designed to fail at a certain load to protect other components or to trigger a safety mechanism.
- It can vary depending on the type of load (static, dynamic, impact, etc.), the type of failure (yielding, buckling, fatigue, etc.), and the type of material (brittle, ductile, composite, etc.) .
- It can be expressed as:

FoS = ultimate strength / working stress

or

FoS = allowable stress / actual stress

or

FoS = load capacity / applied load

Unit 2 - Introduction to IC Engines and Electric Vehicles

Learning Objectives

- To understand the basic principles and components of internal combustion engines (IC engines) and electric vehicles (EVs).
- To compare the advantages and disadvantages of IC engines and EVs in terms of performance, efficiency, emissions, and environmental impact.
- To identify the main types and applications of IC engines and EVs.

IC Engines

- An IC engine is a type of engine that converts the chemical energy of a fuel into mechanical energy by burning it inside a combustion chamber.
- The mechanical energy is then used to move a piston, which in turn drives a crankshaft, a flywheel, and a transmission system to power the wheels of a vehicle.
- There are two kinds of IC engines currently in production: the spark ignition gasoline engine and the compression ignition diesel engine.
- Most of these are four-stroke cycle engines, meaning four piston strokes are needed to complete a cycle: intake, compression, power, and exhaust.
- The main components of an IC engine are: cylinder, piston, connecting rod, crankshaft, valves, spark plug or injector, fuel system, cooling system, lubrication system, and exhaust system.

Electric Vehicles

- An electric vehicle is a type of vehicle that uses an electric motor to convert electrical energy into mechanical energy, or motion, to power the wheels of a vehicle.
- The electrical energy can come from different sources, such as batteries, fuel cells, solar panels, or the grid.
- The main components of an electric vehicle are: electric motor, battery, power electronics, transmission system, and charging system.
- There are different types of electric vehicles, such as battery electric vehicles (BEVs), plug-in hybrid electric vehicles (PHEVs), hybrid electric vehicles (HEVs), and fuel cell electric vehicles (FCEVs).

Comparison of IC Engines and Electric Vehicles

- IC engines have some advantages over electric vehicles, such as: higher range, faster refueling, lower initial cost, and more availability of fuel stations and service centers.
- However, IC engines also have some disadvantages, such as: lower efficiency, higher emissions, higher noise, higher maintenance, and more

dependence on fossil fuels.

- Electric vehicles have some advantages over IC engines, such as: higher efficiency, lower emissions, lower noise, lower maintenance, and more use of renewable energy sources.
- However, electric vehicles also have some disadvantages, such as: lower range, slower charging, higher initial cost, and less availability of charging stations and service centers.

IC Engine

An IC engine is a machine that transforms the energy released from the combustion of fuel into mechanical energy. The motion of the piston is then converted into rotary motion by a crankshaft, which can be used to drive a wide range of machines and vehicles.

Classification of IC Engines

IC engines can be classified according to different criteria, such as:

- The type of fuel used: gas, petrol, kerosene, diesel, etc.
- The working cycle: Otto, Diesel, Dual, etc.
- The speed of the engine: low, medium, high, etc.
- The number of cylinders: single, twin, multi, etc.
- The arrangement of cylinders: inline, V, W, radial, etc.
- The valve arrangement: overhead, side, sleeve, etc.
- The cooling system: air, water, oil, etc.
- The ignition system: spark, compression, etc.
- The fuel injection system: carburetor, direct, indirect, etc.

Components of IC Engines

The main components of an IC engine are:

- Cylinder: It is the cylindrical chamber in which the piston moves and the combustion takes place.
- Piston: It is a cylindrical part that fits inside the cylinder and transmits the force of the combustion to the crankshaft.
- Connecting rod: It is a rod that connects the piston to the crankshaft and converts the reciprocating motion of the piston into rotary motion of the crankshaft.
- Crankshaft: It is a shaft that rotates with the help of the connecting rod and delivers the power to the load.
- Cylinder head: It is the part that covers the top of the cylinder and contains the valves, spark plug, injector, etc.
- Valves: They are the devices that control the flow of air and fuel into the cylinder and the flow of exhaust gases out of the cylinder.

- Spark plug: It is a device that produces a spark to ignite the air-fuel mixture in the cylinder of a spark ignition engine.
- Injector: It is a device that injects the fuel into the cylinder of a compression ignition engine.
- Carburetor: It is a device that mixes the air and fuel in a proper ratio before sending it to the cylinder of a spark ignition engine.
- Air filter: It is a device that filters the air before it enters the engine to remove dust and dirt particles.
- Fuel filter: It is a device that filters the fuel before it enters the engine to remove impurities and water.
- Exhaust manifold: It is a pipe that collects the exhaust gases from the cylinders and sends them to the muffler and the tail pipe.
- Muffler: It is a device that reduces the noise of the exhaust gases by absorbing the sound waves.
- Tail pipe: It is a pipe that releases the exhaust gases into the atmosphere.

Working Principle of IC Engines

The working principle of an IC engine is based on the conversion of chemical energy of the fuel into mechanical energy by the process of combustion. The combustion takes place inside the cylinder at a high temperature and pressure, which pushes the piston downwards. The piston then moves the connecting rod, which rotates the crankshaft. The crankshaft then delivers the power to the load.

The working cycle of an IC engine consists of four strokes: intake, compression, power, and exhaust. These strokes are repeated in a cyclic manner to produce continuous power output. The following diagram shows the four strokes of an IC engine:

Four strokes of an IC engine

- Intake stroke: The piston moves from the top dead center (TDC) to the bottom dead center (BDC), creating a vacuum in the cylinder. The intake valve opens and the air-fuel mixture enters the cylinder through the intake manifold.
- Compression stroke: The piston moves from the BDC to the TDC, compressing the air-fuel mixture in the cylinder. The intake valve closes and the compression ratio increases. The compression ratio is the ratio of the volume of the cylinder when the piston is at the BDC to the volume of the cylinder when the piston is at the TDC.
- Power stroke: The piston is at the TDC and the air-fuel mixture is ignited by the spark plug (in a spark ignition engine) or by the high temperature and pressure (in a compression ignition engine). The combustion releases a large amount of heat and pressure, which pushes the piston downwards with a great force. The power stroke

Basic definition of engine and components

- An engine is a machine that converts one or more forms of energy into mechanical energy that turns the road wheels.
- The most common type of engine in automobiles is the internal combustion engine, which burns fuel and air inside a cylinder to produce power.
- The essential parts of an internal combustion engine include the block, cylinder head, valves, pistons, and piston rings .
- The block is the main structure of the engine that contains the cylinders and the crankshaft.
- The cylinder head is the part that covers the top of the cylinders and houses the valves that let air and fuel mixture into the engine and exhaust gases out of the engine .
- The valves are controlled by a camshaft that opens and closes them at the right time for optimal power production .
- The pistons are cylindrical parts that move up and down inside the cylinders and compress the air and fuel mixture before ignition .
- The piston rings are metal rings that seal the gap between the pistons and the cylinder walls and prevent leakage of gas and oil .
- Other important parts of an internal combustion engine include the crankshaft, connecting rods, flywheel, spark plugs, fuel injectors, oil pump, water pump, and timing belt or chain .
- The crankshaft is the part that converts the linear motion of the pistons into rotational motion that drives the wheels .
- The connecting rods are rods that connect the pistons to the crankshaft and transmit the force of the pistons to the crankshaft .
- The flywheel is a heavy metal disc that is attached to the crankshaft and helps to smooth out the fluctuations in the engine speed .
- The spark plugs are devices that create a spark to ignite the air and fuel mixture in the cylinders .
- The fuel injectors are devices that spray fuel into the cylinders in a precise amount and timing .
- The oil pump is a device that circulates oil throughout the engine to lubricate and cool the moving parts .
- The water pump is a device that circulates coolant throughout the engine to prevent overheating .
- The timing belt or chain is a device that synchronizes the movement of the camshaft and the crankshaft and ensures that the valves open and close at the correct time .

Hello, I am Sydney, your AI assistant. I can help you with your query. Here are some notes on the construction and working of two stroke and four stroke SI and CI engines.

Construction and Working of Two Stroke and Four Stroke SI and CI Engines

Two Stroke Engine

- A two stroke engine is an internal combustion engine that completes one power cycle in two piston strokes or one crankshaft revolution.
- A two stroke engine uses piston-controlled ports instead of valves which makes its construction less complex than the four stroke engine.
- A two stroke engine uses the 'total loss lubrication' method to lubricate the engine components in which the oil is previously added to the fuel. Thus the oil directly enters the engine along with the fuel.
- A two stroke engine can be classified into two types: petrol engine and diesel engine.

Two Stroke Petrol Engine

- A two stroke petrol engine works on the principle of spark ignition. It uses a mixture of air and fuel as the working medium.
- A two stroke petrol engine has the following main components: cylinder, piston, spark plug, inlet port, transfer port, exhaust port, crankcase, and crankshaft.
- A two stroke petrol engine completes one power cycle in the following two strokes:

Suction-Compression Stroke

- In this stroke, the piston moves from the bottom dead center (BDC) to the top dead center (TDC), compressing the fresh charge in the crankcase and the residual gases in the cylinder.
- As the piston moves up, it uncovers the inlet port and allows the fresh charge to enter the crankcase through the carburetor.
- At the same time, the piston covers the transfer port and the exhaust port, sealing the cylinder from the crankcase.
- Near the end of this stroke, the spark plug ignites the compressed charge in the cylinder, causing a rapid increase in pressure and temperature.

Power-Exhaust Stroke

- In this stroke, the piston moves from the TDC to the BDC, expanding the burnt gases and producing useful work on the piston.
- As the piston moves down, it uncovers the exhaust port and allows the burnt gases to escape through the exhaust manifold.
- At the same time, the piston also uncovers the transfer port and allows the fresh charge from the crankcase to enter the cylinder through the deflector on the piston crown.

- The fresh charge helps to scavenge the remaining burnt gases from the cylinder and prepares it for the next cycle.

Two Stroke Diesel Engine

- A two stroke diesel engine works on the principle of compression ignition. It uses only air as the working medium and injects fuel at the end of the compression stroke.
- A two stroke diesel engine has the following main components: cylinder, piston, fuel injector, inlet port, transfer port, exhaust port, crankcase, and crankshaft.
- A two stroke diesel engine completes one power cycle in the following two strokes:

Suction-Compression Stroke

- In this stroke, the piston moves from the BDC to the TDC, compressing the air in the crankcase and the cylinder.
- As the piston moves up, it uncovers the inlet port and allows the fresh air to enter the crankcase through the air filter.
- At the same time, the piston covers the transfer port and the exhaust port, sealing the cylinder from the crankcase.
- Near the end of this stroke, the fuel injector sprays the fuel into the cylinder, where it mixes with the hot compressed air and ignites spontaneously, causing a rapid increase in pressure and temperature.

Power-Exhaust Stroke

- In this stroke, the piston moves from the TDC to the BDC, expanding the burnt gases and producing useful work on the piston.
- As the piston moves down, it uncovers the exhaust port and allows the burnt gases to escape through the exhaust manifold.
- At the same time, the piston also uncovers the transfer port and allows the fresh air from the crankcase to enter the cylinder through the deflector on the piston crown.
- The fresh air helps to scavenge the remaining burnt gases from the cylinder and prepares it for the next cycle.

Four Stroke Engine

- A four stroke engine is an internal combustion engine that completes one power cycle in four piston strokes or two crankshaft revolutions.
- A four stroke engine uses valves to control the flow of the working medium into and out of the cylinder. It also uses a separate lubrication system to lubricate the engine components.
- A four stroke engine can be classified into two types: petrol engine and diesel engine.

Four Stroke Petrol Engine

- A four stroke petrol engine

Merits and Demerits of IC Engines and Electric Vehicles

IC Engines

Merits

- IC engines have a higher power-to-weight ratio than electric motors, which means they can deliver more power for a given size and weight.
- IC engines can use a variety of fuels, such as gasoline, diesel, ethanol, biodiesel, natural gas, etc., which are widely available and relatively cheap.
- IC engines can operate in different weather conditions and altitudes, and do not require special infrastructure or charging stations.
- IC engines have a longer range and faster refueling time than electric vehicles, which means they can travel longer distances without stopping.

Demerits

- IC engines produce harmful emissions, such as carbon dioxide, nitrogen oxides, particulate matter, etc., which contribute to air pollution, climate change, and health problems.
- IC engines have a lower efficiency than electric motors, which means they waste more energy as heat and noise.
- IC engines require more maintenance and servicing than electric motors, which means they have higher operating costs and lower reliability.
- IC engines are dependent on fossil fuels, which are non-renewable and subject to price fluctuations and geopolitical conflicts.

Electric Vehicles

Merits

- Electric vehicles run on electric motors, which have a higher efficiency than IC engines, which means they use less energy and produce less emissions.
- Electric vehicles are quieter and smoother than IC vehicles, which means they offer a more comfortable and relaxing driving experience.
- Electric vehicles have a lower maintenance and servicing cost than IC vehicles, which means they have higher reliability and longer lifespan.
- Electric vehicles can use renewable energy sources, such as solar, wind, hydro, etc., which are clean and sustainable.

Demerits

- Electric vehicles have a lower power-to-weight ratio than IC engines, which means they have lower performance and acceleration.
- Electric vehicles have a limited range and longer charging time than IC vehicles, which means they need more frequent stops and planning.
- Electric vehicles require special infrastructure and charging stations, which are not widely available and may be expensive.
- Electric vehicles have a higher upfront cost than IC vehicles, which means they need more initial investment and incentives.

Scavenging Process

- Scavenging is the process of replacing the exhaust gas in a cylinder of an internal combustion engine with the fresh air/fuel mixture (or fresh air, in the case of direct-injection engines) for the next cycle.
- Scavenging is necessary for the engine's proper fuel combustion and better power output .
- Scavenging occurs in ic engine during the overlapping of valves (Opens at the same time) in which burnt gases are released out of the cylinder with the entering of fresh charge inside the engine at the same time.
- There are three types of scavenging used in two stroke IC engines:
 - Uniflow scavenging: The fresh charge enters from one end of the cylinder and the exhaust gas exits from the other end. This type of scavenging has the highest efficiency and the least short-circuiting of the fresh charge.
 - Cross-flow scavenging: The fresh charge enters from one side of the cylinder and the exhaust gas exits from the opposite side. This type of scavenging has a lower efficiency and a higher short-circuiting of the fresh charge than uniflow scavenging .
 - Loop or reverse scavenging: The fresh charge enters from the bottom of the cylinder and the exhaust gas exits from the top. This type of scavenging has the lowest efficiency and the highest short-circuiting of the fresh charge among the three types .
- The scavenging efficiency is the ratio of the mass of fresh charge trapped in the cylinder to the mass of fresh charge supplied to the cylinder. It is a measure of how well the scavenging process removes the exhaust gas and fills the cylinder with the fresh charge.

Difference between two-stroke and four stroke IC engines and SI and CI Engines

Two-stroke and four stroke IC engines

- Two-stroke and four stroke IC engines are types of internal combustion engines that use the combustion of fuel and air to produce mechanical power.
- The main difference between them is the number of strokes or movements of the piston in one cycle of operation.
- A two-stroke engine completes one cycle in two strokes or one revolution of the crankshaft, while a four-stroke engine completes one cycle in four strokes or two revolutions of the crankshaft .
- The four strokes of a four-stroke engine are: intake, compression, power, and exhaust. During the intake stroke, the piston moves down and draws air and fuel into the cylinder. During the compression stroke, the piston moves up and compresses the air-fuel mixture. During the power stroke, the spark plug ignites the compressed mixture and the piston is pushed down by the expanding gases. During the exhaust stroke, the piston moves up and pushes the burnt gases out of the cylinder.
- The two strokes of a two-stroke engine are: power and exhaust. During the power stroke, the piston moves down and compresses the air-fuel mixture in the crankcase, while the spark plug ignites the mixture in the cylinder. The expanding gases push the piston down and also open the exhaust port. During the exhaust stroke, the piston moves up and pushes the burnt gases out of the cylinder, while also opening the intake port. The fresh air-fuel mixture from the crankcase enters the cylinder and the cycle repeats.
- Some advantages of two-stroke engines are: simpler design, lighter weight, higher power-to-weight ratio, and lower cost . Some disadvantages are: higher fuel consumption, higher emissions, lower durability, and higher noise .
- Some advantages of four-stroke engines are: lower fuel consumption, lower emissions, higher durability, and lower noise . Some disadvantages are: more complex design, heavier weight, lower power-to-weight ratio, and higher cost .

SI and CI Engines

- SI and CI engines are types of internal combustion engines that differ in the way they ignite the air-fuel mixture in the cylinder.
- SI stands for spark ignition, which means that the air-fuel mixture is ignited by a spark plug. CI stands for compression ignition, which means that the air-fuel mixture is ignited by the high temperature and pressure created by the compression stroke.
- SI engines use a homogeneous mixture of air and fuel, which is drawn into the cylinder during the intake stroke. CI engines use a heterogeneous

mixture of air and fuel, which is injected into the cylinder near the end of the compression stroke.

- SI engines can use gasoline, ethanol, or natural gas as fuel. CI engines can use diesel, biodiesel, or vegetable oil as fuel.
- Some advantages of SI engines are: higher power output, smoother operation, and easier starting. Some disadvantages are: lower thermal efficiency, higher fuel consumption, and higher emissions of carbon monoxide and unburned hydrocarbons.
- Some advantages of CI engines are: higher thermal efficiency, lower fuel consumption, and lower emissions of carbon monoxide and unburned hydrocarbons. Some disadvantages are: lower power output, rougher operation, and harder starting.

Electric vehicles and hybrid vehicles

- Electric vehicles (EVs) are vehicles that use an electric motor and a battery pack as the main source of propulsion. EVs do not have an internal combustion engine (ICE) or a fuel tank, and they do not emit tailpipe emissions. EVs can be recharged by plugging into an external power source, such as a wall outlet or a charging station .
- Hybrid vehicles are vehicles that use a combination of an ICE and an electric motor to provide propulsion. Hybrid vehicles have both a fuel tank and a battery pack, and they can switch between using gasoline and electricity depending on the driving conditions. Hybrid vehicles can reduce fuel consumption and emissions compared to conventional vehicles, but they still rely on fossil fuels .
- Plug-in hybrid electric vehicles (PHEVs) are a type of hybrid vehicle that can be plugged into an external power source to recharge their battery pack. PHEVs have a larger battery capacity and a longer electric range than conventional hybrid vehicles, and they can operate as pure EVs when the battery is fully charged. PHEVs can offer the benefits of both EVs and hybrid vehicles, but they are more expensive and complex to maintain .
- Some of the advantages of EVs and hybrid vehicles are:
 - They can reduce greenhouse gas emissions and air pollution by using less or no gasoline .
 - They can lower fuel costs and increase energy security by using electricity from various sources, such as renewable energy .
 - They can improve performance and efficiency by using regenerative braking, which recovers kinetic energy and stores it in the battery .
 - They can offer a smoother and quieter ride than conventional vehicles, as they have fewer moving parts and less engine noise .
- Some of the challenges and limitations of EVs and hybrid vehicles are:
 - They have a higher upfront cost than conventional vehicles, as they require more advanced technology and materials .
 - They have a limited driving range and availability of charging in-

frastructure, especially for EVs, which can cause range anxiety and inconvenience for drivers .

- They have a lower resale value and a shorter battery life than conventional vehicles, as they are subject to degradation and obsolescence over time .
- They have a higher environmental impact in terms of battery production and disposal, as they use rare and toxic materials that can cause pollution and resource depletion .

Components of an EV

An electric vehicle (EV) is a vehicle that uses one or more electric motors for propulsion. Unlike conventional vehicles that use an internal combustion engine (ICE), EVs do not emit tailpipe emissions and can be powered by renewable energy sources. EVs have several advantages over ICE vehicles, such as lower operating and maintenance costs, higher efficiency, and reduced noise and vibration.

The main components of an EV are :

- **Traction battery pack:** This is the main source of energy for the EV. It consists of a series of rechargeable cells that store and deliver electricity to the electric motor. The battery pack can be charged through an external power supply, such as a charging station or a household outlet. The battery pack also provides electricity to power the vehicle accessories, such as lights, air conditioning, and radio. The battery pack is usually located under the floor or in the trunk of the vehicle.
- **DC-DC converter:** This is a device that converts the high-voltage direct current (DC) from the battery pack to a lower-voltage DC for the electric motor and the auxiliary battery. The DC-DC converter also regulates the voltage to maintain a constant power output.
- **Electric motor:** This is the main component of the EV that converts the electrical energy from the battery pack into mechanical energy that drives the wheels. The electric motor can be either AC (alternating current) or DC, depending on the type of EV. The electric motor can also act as a generator, by recovering the kinetic energy from the braking and coasting of the vehicle and converting it back into electrical energy that is stored in the battery pack. This process is called regenerative braking and it helps to extend the range and efficiency of the EV.
- **Power inverter:** This is a device that converts the DC from the battery pack or the DC-DC converter to AC for the electric motor. The power inverter also controls the speed and torque of the electric motor by varying the frequency and amplitude of the AC.
- **Charge port:** This is the interface that allows the EV to connect to an

external power supply for charging the battery pack. The charge port can be either AC or DC, depending on the type of charging station. The charge port can also communicate with the charging station to exchange information, such as the battery state of charge, the charging time, and the charging cost.

- **Onboard charger:** This is a device that converts the AC from the charge port to DC for the battery pack. The onboard charger also regulates the current and voltage to optimize the charging process and protect the battery pack from overcharging or overheating.
- **Controller:** This is the central unit that manages the operation of the EV. The controller receives inputs from the driver, such as the accelerator pedal, the brake pedal, and the steering wheel, and sends commands to the electric motor, the power inverter, the DC-DC converter, and the onboard charger. The controller also monitors the performance and status of the EV, such as the battery state of charge, the motor speed and temperature, and the vehicle speed and distance.
- **Auxiliary battery:** This is a secondary battery that provides electricity to power the vehicle accessories when the EV is not in operation or when the traction battery pack is low. The auxiliary battery is usually a 12-volt lead-acid battery that is charged by the DC-DC converter. The auxiliary battery is similar to the battery used in ICE vehicles.

EV batteries

- EV batteries are the devices that store electrical energy and power the electric motors in electric vehicles (EVs).
- EV batteries are usually rechargeable and can be charged from external sources or from regenerative braking, which converts kinetic energy into electricity when the vehicle slows down.
- The most common type of EV battery is lithium-ion, which uses lithium metal and other materials as electrodes and an electrolyte solution as the medium for ion transfer.
- Lithium-ion batteries have high energy density, meaning they can store a lot of energy in a small volume and weight. They also have low self-discharge rate, meaning they can retain their charge for a long time when not in use.
- Other types of EV batteries include lead-acid, nickel-cadmium, nickel-metal hydride, zinc-air, and sodium nickel chloride, but they are less common due to their lower performance, higher cost, or environmental issues.
- EV batteries have several advantages over internal combustion engines (ICEs), such as lower emissions, higher efficiency, lower noise, and lower maintenance.

- However, EV batteries also have some challenges, such as high cost, limited range, long charging time, degradation over time, and recycling issues.
- EV batteries are constantly being improved by research and innovation, such as developing new materials, increasing capacity, reducing weight, enhancing safety, and lowering cost.
- Some of the emerging technologies for EV batteries include solid-state batteries, which use solid electrolytes instead of liquid ones, and wireless charging, which uses electromagnetic induction to transfer power without physical contact.

Chargers for Electric Vehicles

- Electric vehicles (EVs) need chargers to replenish their batteries and extend their driving range.
- There are different types and levels of chargers for EVs, depending on the power source, the charging speed, and the connector type.
- The main types of chargers for EVs are:
 - Level 1 chargers: These are the most common and basic chargers that can plug into a standard 120-volt (120V) AC wall outlet. They usually come with the vehicle at purchase and deliver about 1.4 kilowatts (kW) of charge, providing 4 miles of driving range per hour of charging. They are suitable for overnight or long-term charging at home or work, but they are not very fast or efficient.
 - Level 2 chargers: These are faster and more efficient chargers that can plug into a 240-volt (240V) AC outlet or a dedicated charging station. They deliver about 3.3 to 19.2 kW of charge, providing 10 to 60 miles of driving range per hour of charging. They are suitable for daily or short-term charging at home, work, or public places. They require a special installation and may cost more than Level 1 chargers.
 - Level 3 chargers: These are the fastest and most powerful chargers that can use a 480-volt (480V) DC current or a high-voltage AC current. They deliver about 50 to 350 kW of charge, providing 180 to 1000 miles of driving range per hour of charging. They are suitable for rapid or emergency charging on highways or long trips. They are also known as DC fast chargers or superchargers. They are not widely available and may be expensive to use.
- The main types of connectors for EV chargers are:
 - SAE J1772: This is the standard connector for Level 1 and Level 2 chargers in the US. It is compatible with most EVs, except Tesla. It

has five pins and a circular shape.

- Tesla: This is the proprietary connector for Tesla vehicles and chargers. It is compatible with Level 1, Level 2, and Level 3 chargers, but only with Tesla vehicles. It has four pins and a flat shape.
- CHAdeMO: This is the standard connector for Level 3 chargers in Japan and some other countries. It is compatible with some EVs, such as Nissan Leaf and Mitsubishi Outlander. It has 10 pins and a D-shape.
- CCS: This is the standard connector for Level 3 chargers in Europe and some other countries. It is compatible with some EVs, such as BMW i3 and Volkswagen e-Golf. It has seven pins and a combined shape of SAE J1772 and DC pins.
- Some of the best home EV chargers for 2023 are:
 - JuiceBox 40 Smart Charging Station: This is a Level 2 charger that can deliver up to 40 amps of charge and has smart features such as Wi-Fi connectivity, app control, voice control, and energy monitoring. It is compatible with all EVs using the SAE J1772 connector. It costs about \$590.
 - Grizzl-E 40 Amp Charger: This is a Level 2 charger that can deliver up to 40 amps of charge and has a rugged design that can withstand outdoor use. It is compatible with all EVs using the SAE J1772 connector. It costs about \$799.
 - EVoCharge Electric Vehicle Charging Station: This is a Level 2 charger that can deliver up to 32 amps of charge and has a sleek design that can fit in any garage. It is compatible with all EVs using the SAE J1772 connector. It costs about \$568.
 - Morec 32 Amp Level 2 Charger: This is a Level 2 charger that can deliver up to 32 amps of charge and has a portable design that can be easily installed and removed. It is compatible with all EV

Drives for IC Engines and Electric Vehicles

- A drive is a system that transfers power from a source to a load, such as a vehicle's wheels.
- There are two main types of drives for vehicles: internal combustion engine (ICE) drives and electric vehicle (EV) drives.
- ICE drives use fuel, such as gasoline or diesel, that combusts inside a combustion chamber with the help of an oxidizer, such as air, to produce

mechanical power. The power is then transmitted to the wheels through a transmission, a clutch, a differential, and an axle. ICE drives are widely used in conventional vehicles, such as cars, trucks, buses, motorcycles, etc.

- EV drives use one or more electric motors that convert electrical energy from a battery or another source into mechanical power. The power is then transmitted to the wheels through a single or multiple gear ratio, a differential, and an axle. EV drives are used in electric vehicles, such as battery electric vehicles (BEVs), plug-in hybrid electric vehicles (PHEVs), hybrid electric vehicles (HEVs), fuel cell electric vehicles (FCEVs), etc.
- The main advantages of EV drives over ICE drives are:
 - Higher efficiency: EV drives can convert more than 90% of the electrical energy into mechanical power, while ICE drives can only convert about 20-30% of the fuel energy into mechanical power. EV drives also have less energy losses due to friction, heat, and noise.
 - Lower emissions: EV drives do not produce tailpipe emissions, such as carbon dioxide, nitrogen oxides, particulate matter, etc., that contribute to air pollution and climate change. EV drives can also reduce greenhouse gas emissions if the electricity is generated from renewable sources, such as solar, wind, hydro, etc.
 - Better performance: EV drives can provide instant torque and acceleration, as well as smoother and quieter operation. EV drives can also enable regenerative braking, which means the electric motor can operate in reverse and recapture some of the kinetic energy of the vehicle and charge the battery, thereby extending the driving range and reducing brake wear.
- The main disadvantages of EV drives over ICE drives are:
 - Higher cost: EV drives are more expensive than ICE drives, mainly due to the high cost of batteries, which account for a large portion of the vehicle's price. EV drives also require more complex and sophisticated control systems and components, such as inverters, converters, chargers, etc.
 - Limited range: EV drives have a shorter driving range than ICE drives, as the battery capacity and performance are affected by various factors, such as temperature, state of charge, driving conditions, etc. EV drives also depend on the availability and accessibility of charging stations, which are still limited in some areas.
 - Longer charging time: EV drives take longer to recharge than ICE drives, which can be refueled in a few minutes. EV drives can take several hours to fully charge, depending on the battery size and the charging power. EV drives also face the risk of battery degradation and capacity loss over time and use.

Transmission and Power Devices for IC Engines and Electric Vehicles

- Transmission is a machine that transmits power from the energy source (electric motor or internal combustion engine) to the wheels of the vehicle.
- Power devices are components that convert, regulate, or control the power flow in the transmission system.
- Some examples of transmission and power devices for IC engines and electric vehicles are:
 - Power split device: A device that can operate the vehicle with electric motor power and the IC engine separately or can also combine the power from both sources. It also acts as a continuously variable transmission (CVT) and eliminates the need for any manual or automatic transmission.
 - Motor: A device that converts electrical energy into mechanical energy. Electric vehicles use electric motors to drive the wheels, while hybrid vehicles use both electric motors and IC engines. Electric motors are more efficient and have fewer moving parts than IC engines.
 - Differential: A device that allows the wheels to rotate at different speeds while receiving power from the transmission. It is essential for smooth turning and cornering of the vehicle. Differentials can be mechanical, electronic, or hydraulic.
 - Voltage regulator: A device that maintains a constant voltage level for the electrical system of the vehicle. It is used to protect the battery, the alternator, and the electronic components from voltage fluctuations. Voltage regulators can be integrated into the alternator or the engine control unit (ECU) of the vehicle.
 - Sensor interface: A device that converts the signals from various sensors (such as temperature, pressure, speed, etc.) into digital or analog signals that can be processed by the ECU. Sensor interfaces can be analog, digital, or mixed-signal devices.
 - Load or motor driver: A device that controls the current or voltage supplied to a load or a motor. It is used to regulate the speed, torque, or direction of the motor or the load. Load or motor drivers can be linear, switching, or brushless devices.

Advantages and Disadvantages of EVs

Electric vehicles (EVs) are vehicles that use electric motors instead of internal combustion engines (ICEs) to propel themselves. EVs can be powered by batteries, fuel cells, or other sources of electricity. EVs have several advantages and disadvantages compared to ICE vehicles, which are discussed below.

Advantages of EVs

- **Environmental impact:** EVs have lower greenhouse gas emissions and air pollution than ICE vehicles, as they do not burn fossil fuels or produce tailpipe emissions . EVs can also use renewable energy sources, such as solar or wind, to charge their batteries, reducing their carbon footprint even further .
- **Energy efficiency:** EVs are more efficient than ICE vehicles, as they convert a higher percentage of the electrical energy from the grid to power the wheels. ICE vehicles lose a lot of energy as heat and noise during the combustion process . EVs also have regenerative braking, which recovers some of the kinetic energy that would otherwise be wasted when slowing down .
- **Cost savings:** EVs have lower operating and maintenance costs than ICE vehicles, as they have fewer moving parts, do not require oil changes or tune-ups, and have lower fuel costs. The cost of electricity is generally cheaper and more stable than the cost of gasoline or diesel . EVs may also qualify for tax credits, rebates, or other incentives from the government or utility companies .
- **Performance:** EVs have instant torque, which means they can accelerate faster and smoother than ICE vehicles. EVs also have lower center of gravity, which improves their handling and stability. EVs are quieter and more comfortable to drive than ICE vehicles, as they do not have engine noise or vibration .

Disadvantages of EVs

- **Recharge points:** EVs have limited range and need to be recharged frequently, which can be inconvenient and time-consuming. The availability and accessibility of charging stations are still limited, especially in rural areas or long-distance routes. EV drivers may face range anxiety, which is the fear of running out of battery before reaching their destination or a charging point .
- **Initial investment:** EVs have higher upfront costs than ICE vehicles, mainly due to the expensive batteries. The batteries also degrade over time and need to be replaced, which adds to the long-term cost of ownership. EVs may not be affordable or cost-effective for some consumers, especially if they do not drive a lot or have access to cheap electricity .
- **Electricity source:** EVs are only as green as the electricity that powers them. If the electricity comes from coal or other fossil fuels, EVs may still contribute to greenhouse gas emissions and air pollution, albeit indirectly. The increased demand for electricity from EVs may also strain the existing grid infrastructure and require more investment in generation and transmission capacity .
- **Driving range and speed:** EVs have lower driving range and top speed than ICE vehicles, as they are limited by the battery capacity and power

output. EVs may not be suitable for long-distance trips or high-speed driving, as they may run out of battery or lose performance. EVs may also face challenges in extreme weather conditions, such as cold or hot temperatures, which can affect the battery performance and efficiency .

- **Variety and design:** EVs have fewer models and options than ICE vehicles, as they are still a niche market and have less competition and innovation. EVs may not meet the preferences or needs of some consumers, who may prefer the design, features, or performance of ICE vehicles. EVs may also have less cargo space or passenger capacity than ICE vehicles, as they need to accommodate the batteries and other components .

: <https://www.roadcartel.com/blog/what-are-the-disadvantages-of-electric-cars>
<https://www.carsguide.com.au/ev/advice/the-pros-and-cons-of-electric-cars->

Hybrid electric vehicles

- A hybrid electric vehicle (HEV) is a type of hybrid vehicle that combines a conventional internal combustion engine (ICE) system with an electric propulsion system (hybrid vehicle drivetrain) .
- The presence of the electric powertrain is intended to achieve either better fuel economy than a conventional vehicle or better performance .
- A hybrid electric vehicle cannot be plugged in to charge the battery. Instead, the battery is charged through regenerative braking and by the internal combustion engine .
- There are different types of hybrid electric vehicles, such as series hybrids, parallel hybrids, series-parallel hybrids, mild hybrids, and plug-in hybrids .
- Series hybrids have only an electric motor to drive the wheels, and the ICE acts as a generator to charge the battery .
- Parallel hybrids have both an ICE and an electric motor connected to the transmission, and they can work together or separately to propel the vehicle .
- Series-parallel hybrids can operate as either series or parallel hybrids depending on the driving conditions .
- Mild hybrids have a small electric motor that assists the ICE, but cannot drive the vehicle by itself .
- Plug-in hybrids (PHEVs) are a combination of gasoline and electric vehicles, so they have a battery, an electric motor, a gasoline tank, and an internal combustion engine. PHEVs use both gasoline and electricity as fuel sources. They can be plugged in to an external power source to charge the battery, and they can also use regenerative braking and the ICE to charge the battery .
- Hybrid electric vehicles have several advantages over conventional vehicles, such as lower emissions, higher fuel efficiency, reduced dependence on fossil

fuels, and improved performance .

- Hybrid electric vehicles also have some challenges, such as higher initial cost, complexity of the hybrid system, battery weight and size, and limited driving range on electric mode .
- Some examples of hybrid electric vehicles are Toyota Prius, Honda Insight, Ford Fusion Hybrid, Hyundai Sonata Hybrid, Kia Niro Plug-In Hybrid, and Chevrolet Volt .

HEV drive train components

A hybrid electric vehicle (HEV) is a type of hybrid vehicle that combines a conventional internal combustion engine (ICE) system with an electric propulsion system (hybrid vehicle drivetrain). The key components in an HEV drive train are :

- **Internal combustion engine:** The primary power source of an HEV is a conventional engine. It can run on gasoline, diesel, or other fuels. The engine provides mechanical power to the wheels or to a generator that produces electrical power.
- **Electric motor:** The secondary power source of an HEV is the electric motor. It converts electrical energy from the battery pack or the generator into mechanical power that drives the wheels. The electric motor can also act as a generator during braking or coasting, recovering kinetic energy and storing it in the battery pack.
- **Battery pack:** A battery pack powers the electric motor. It stores electrical energy that can be used to supplement or replace the engine power. The battery pack can be recharged by the engine, the electric motor, or an external source (in plug-in hybrid vehicles).
- **Generator:** It is an essential component found in the series hybrid vehicle. It converts mechanical power from the engine into electrical power that can be used by the electric motor or stored in the battery pack. The generator allows the engine to operate at its optimal efficiency and emissions level, regardless of the vehicle speed or load.
- **Transmission:** Typically, hybrid vehicles use conventional transmissions similar to petrol or diesel cars. The transmission transfers the power from the engine and/or the electric motor to the wheels. Depending on the type of hybrid vehicle, the transmission can be a manual, automatic, continuously variable, or power-split device.
- **Control board:** It is the brain of the hybrid vehicle. It monitors and regulates the operation of the drive train components. It determines the optimal power split between the engine and the electric motor, based on the driver's input and the vehicle's condition. It also controls the charging and discharging of the battery pack, the regenerative braking, and the start-stop function of the engine.

Advantages of HV

HV stands for high voltage, which is a term used to describe electrical power transmission systems that operate at voltages above 100 kV (kilovolts). HV systems are used to transmit large amounts of power over long distances, such as from power plants to cities or across continents. HV systems can also be used to connect different power grids or to integrate renewable energy sources.

Some of the advantages of HV systems are:

- **Reduced power loss:** HV systems reduce the power loss due to the resistance of the transmission wires, as the power loss is proportional to the square of the current. By increasing the voltage, the current is reduced for the same power, and hence the power loss is reduced. For example, a power loss of only 3% for every 1000 km can be achieved with HV systems, depending on the system construction and voltage level.
- **Reduced wire size and cost:** HV systems reduce the size and cost of the transmission wires, as the wire size is inversely proportional to the voltage. By increasing the voltage, the wire size can be reduced for the same current, and hence the wire cost can be reduced. For example, a HV system can use a thin conductor to transmit bulk power over a long distance.
- **Reduced space and weight:** HV systems reduce the space and weight of the transmission equipment, such as generators, motors, transformers, and cables, as they are directly related to the current. By increasing the voltage, the current is reduced for the same power, and hence the space and weight of the equipment can be reduced. For example, a HV system can reduce the overall system size and weight compared to the same power low voltage system.
- **Improved voltage regulation:** HV systems improve the voltage regulation and reduce the voltage drop, as the voltage drop is proportional to the current and the resistance of the transmission wires. By increasing the voltage, the current is reduced for the same power, and hence the voltage drop is reduced. For example, a HV system can maintain a constant voltage at the load end, regardless of the distance from the source.
- **Less fossil fuel dependency:** HV systems can reduce the dependency on fossil fuels, as they can integrate renewable energy sources, such as wind, solar, hydro, and geothermal, which are often located far away from the load centers. By using HV systems, the renewable energy can be transmitted efficiently and reliably to the consumers, and hence the fossil fuel consumption and greenhouse gas emissions can be reduced. For example, a hybrid vehicle can use a high voltage battery pack and electric motor(s) for propulsion, and hence rely less on fossil fuels than a standard gas car.

Unit 3 - Introduction to Refrigeration and Air-Conditioning

- Refrigeration and air-conditioning are processes that involve the transfer of heat from a lower-temperature region to a higher-temperature region, using a substance called a refrigerant as the working medium.
- Refrigeration is used to create and maintain low temperatures for various purposes, such as food preservation, industrial cooling, cryogenics, etc.
- Air-conditioning is used to provide thermal comfort and improve the quality of indoor air, by controlling the temperature, humidity, ventilation, and filtration of the air.
- The basic principle of refrigeration and air-conditioning is the same: a refrigerant undergoes a cycle of phase changes (evaporation and condensation) and pressure changes (compression and expansion) to absorb heat from a cold space and reject it to a hot space.
- The main components of a refrigeration or air-conditioning system are: an evaporator, where the refrigerant absorbs heat from the cold space and evaporates; a compressor, where the refrigerant is compressed to a high pressure and temperature; a condenser, where the refrigerant releases heat to the hot space and condenses; and an expansion device, where the refrigerant is expanded to a low pressure and temperature.
- The performance of a refrigeration or air-conditioning system is measured by the coefficient of performance (COP), which is the ratio of the useful heat transfer (at the evaporator or the condenser) to the work input (by the compressor) .
- The COP depends on the type of refrigerant, the operating conditions, and the design of the system. The higher the COP, the more efficient the system is .
- There are different types of refrigeration and air-conditioning systems, such as vapor compression, vapor absorption, thermoelectric, air cycle, etc., each with its own advantages and disadvantages.

Refrigeration

Refrigeration is the process of removing heat from a substance or a space and transferring it to another place where it can be rejected or used. Refrigeration is used to lower and/or maintain the temperature of a substance or a space below the ambient temperature. Refrigeration is an artificial or human-made cooling method.

Some of the uses of refrigeration are:

- To preserve food and beverages by slowing down the growth of microorganisms and chemical reactions.
- To provide thermal comfort by cooling air in buildings, vehicles, and other enclosed spaces.
- To enable industrial processes that require low temperatures, such as liq-

uefaction of gases, cryogenic applications, ice making, etc.

- To improve the efficiency and performance of some machines and devices, such as gas turbines, heat pumps, etc.

Types of Refrigeration

There are many types of refrigeration systems, but they can be broadly classified into two categories based on the working principle: **vapor compression refrigeration** and **vapor absorption refrigeration**.

Vapor Compression Refrigeration

Vapor compression refrigeration is the most common and widely used type of refrigeration system. It works on the principle of compressing a refrigerant (a substance that can change its phase from liquid to vapor and vice versa) in a closed cycle and using it to absorb heat from a low-temperature source and reject it to a high-temperature sink.

The main components of a vapor compression refrigeration system are:

- **Compressor:** It is a device that compresses the low-pressure refrigerant vapor coming from the evaporator and raises its pressure and temperature.
- **Condenser:** It is a heat exchanger that cools down the high-pressure refrigerant vapor coming from the compressor and condenses it into a liquid.
- **Expansion valve:** It is a device that reduces the pressure and temperature of the high-pressure refrigerant liquid coming from the condenser and expands it into a low-pressure liquid-vapor mixture.
- **Evaporator:** It is a heat exchanger that absorbs heat from the low-temperature source (such as a refrigerator or an air conditioner) and evaporates the low-pressure refrigerant liquid-vapor mixture coming from the expansion valve into a vapor.

The following diagram shows the schematic of a vapor compression refrigeration system and the corresponding pressure-enthalpy (P-h) diagram of the refrigerant cycle.

Vapor compression refrigeration system and P-h diagram

Vapor Absorption Refrigeration

Vapor absorption refrigeration is another type of refrigeration system that works on the principle of absorbing a refrigerant vapor (usually ammonia) in a liquid (usually water) and using it to absorb heat from a low-temperature source and reject it to a high-temperature sink.

The main components of a vapor absorption refrigeration system are:

- **Generator:** It is a device that heats up the refrigerant-absorbent solution (usually ammonia-water) coming from the absorber and separates the refrigerant vapor from the absorbent liquid.
- **Condenser:** It is a heat exchanger that cools down the refrigerant vapor coming from the generator and condenses it into a liquid.
- **Expansion valve:** It is a device that reduces the pressure and temperature of the refrigerant liquid coming from the condenser and expands it into a low-pressure liquid-vapor mixture.
- **Evaporator:** It is a heat exchanger that absorbs heat from the low-temperature source (such as a refrigerator or an air conditioner) and evaporates the refrigerant liquid-vapor mixture coming from the expansion valve into a vapor.
- **Absorber:** It is a device that absorbs the refrigerant vapor coming from the evaporator and mixes it with the absorbent liquid coming from the generator.

The following diagram shows the schematic of a vapor absorption refrigeration system and the corresponding temperature-entropy (T-s) diagram of the refrigerant cycle.

Vapor absorption refrigeration system and T-s diagram

Classification of Refrigerants

Refrigerants are the substances that are used as the working fluid in refrigeration systems. They can be classified into different categories based on their chemical composition, thermodynamic properties, environmental impact, safety, and availability.

Some of the common categories of refrigerants are:

- **Primary refrigerants:** They are the refrigerants that directly

Refrigerating Effect

- Refrigerating effect is the amount of heat transferred by one kg of refrigerant as it circulates in the refrigeration system.
- Refrigerating effect is also the heat absorbed in the evaporator per unit mass of refrigerant.
- Refrigerating effect is determined by the difference in enthalpy of the refrigerant vapor leaving the evaporator and the refrigerant liquid just upstream of the expansion valve at the evaporator.
- Refrigerating effect is the measure of the cooling capacity of a refrigeration system.
- Refrigerating effect is the opposite of the heat output of a heat engine.
- Refrigerating effect can be expressed in kJ/kg or Btu/lb of refrigerant .

- Refrigerating effect can be increased by increasing the temperature difference between the refrigerant and the medium to be cooled, or by decreasing the pressure difference between the condenser and the evaporator.

Ton of Refrigeration

- A ton of refrigeration (TR or TOR), also called a refrigeration ton (RT), is a unit of power used in some countries (especially in North America) to describe the heat-extraction capacity of refrigeration and air conditioning equipment .
- It is defined as the amount of heat removed from a given space in one hour by a refrigeration system that can freeze one short ton (2000 lb or 907 kg) of water at 32 °F (0 °C) in 24 hours.
- One ton of refrigeration is equal to 12,000 British thermal units per hour (BTU/h) or 3.517 kilowatts (kW).
- The origin of the term comes from the ice-making industry in the 19th century, when ice was harvested from frozen lakes and rivers and stored in ice houses for later use. The rate of heat transfer required to produce one ton of ice in 24 hours was used as a benchmark for the cooling capacity of refrigeration machines.
- The ton of refrigeration is still widely used in the HVAC industry, especially for commercial and industrial applications. However, it is not an SI unit and is not recognized by the International System of Units (SI). The SI unit of cooling capacity is the watt (W).

Coefficient of Performance for Refrigeration and Air-Conditioning

- The coefficient of performance (COP) is a measure of the efficiency of a refrigeration or air-conditioning system. It is defined as the ratio of the useful cooling or heating effect to the work input required by the system.

- The COP of a refrigeration system is given by:

$$\text{COP} = Q_c / W$$

where Q_c is the heat removed from the cold reservoir (inside the refrigerator) and W is the work done by the compressor to remove the heat.

- The COP of an air-conditioning system is given by:

$$\text{COP} = Q_h / W$$

where Q_h is the heat delivered to the hot reservoir (inside the room) and W is the work done by the compressor to deliver the heat.

- The COP of a system depends on the temperature difference between the hot and cold reservoirs, the type of refrigerant used, and the design of the system components.
- The higher the COP, the more efficient the system is, and the lower the energy consumption and operating costs are.
- The typical COP values for refrigeration systems range from 2.6 to 3.0 for cutting and preparation rooms, 3.0 to 4.0 for cold storage rooms, and 4.0 to 6.0 for freezers.
- The typical COP values for air-conditioning systems range from 2.0 to 4.0 for window units, 3.0 to 5.0 for split systems, and 4.0 to 6.0 for central systems.
- The COP of a system can be improved by using more efficient compressors, evaporators, and condensers, reducing heat losses and leakages, and maintaining proper refrigerant charge and pressure.

Methods of Refrigeration

Refrigeration is the process of removing heat from a substance, space, or system and transferring it to another substance, space, or system. Refrigeration has many applications, such as preserving food, cooling air, and producing ice. There are different methods of refrigeration, depending on the working principle, the refrigerant used, and the desired temperature level. Some of the common methods of refrigeration are:

- **Ice refrigeration:** This method uses ice as the refrigerant, which absorbs heat from the substance or space to be cooled and melts into water. Ice refrigeration is one of the oldest and simplest methods of refrigeration, but it has low efficiency and requires a large amount of ice. Ice refrigeration is mainly used for domestic purposes, such as cooling drinks and food items.
- **Dry ice refrigeration:** This method uses dry ice, which is solid carbon dioxide, as the refrigerant. Dry ice sublimates (changes from solid to gas) at a low temperature and pressure, and absorbs a large amount of heat in the process. Dry ice refrigeration is used for low-temperature applications, such as freezing food, transporting biological samples, and creating special effects.
- **Evaporative refrigeration:** This method uses the evaporation of a liquid, such as water or alcohol, as the refrigeration process. Evaporation is a natural phenomenon that occurs when a liquid changes into vapor and absorbs heat from its surroundings. Evaporative refrigeration is a cheap and eco-friendly method of refrigeration, but it has low efficiency and de-

depends on the humidity and temperature of the environment. Evaporative refrigeration is used for cooling buildings, vehicles, and animals .

- **Liquid gas refrigeration:** This method uses a liquid gas, such as liquid nitrogen or liquid helium, as the refrigerant. Liquid gas refrigeration is based on the Joule-Thomson effect, which states that a gas cools down when it expands through a valve or a nozzle. Liquid gas refrigeration is used for very low-temperature applications, such as cryogenics, superconductivity, and space exploration.
- **Gas throttling refrigeration:** This method uses a gas, such as air or ammonia, as the refrigerant. Gas throttling refrigeration is also based on the Joule-Thomson effect, but it uses a compressor and a condenser to create a closed cycle of refrigeration. Gas throttling refrigeration is one of the most widely used methods of refrigeration, as it has high efficiency and can achieve a wide range of temperatures. Gas throttling refrigeration is used for air conditioning, industrial refrigeration, and heat pumps .
- **Air expansion refrigeration:** This method uses air as the refrigerant, and it is similar to gas throttling refrigeration, except that it uses an expander instead of a valve or a nozzle to expand the air. Air expansion refrigeration has higher efficiency than gas throttling refrigeration, as it recovers some of the work done by the compressor. Air expansion refrigeration is used for aircraft refrigeration, liquefaction of gases, and low-temperature research.
- **Vapor compression refrigeration:** This method uses a vapor, such as water or refrigerant, as the refrigerant. Vapor compression refrigeration is the most common method of refrigeration, and it consists of four main components: a compressor, a condenser, an expansion valve, and an evaporator. Vapor compression refrigeration works by compressing the vapor, condensing it into a liquid, expanding it through a valve, and evaporating it into a vapor again, while absorbing heat from the substance or space to be cooled. Vapor compression refrigeration is used for domestic refrigerators, commercial freezers, and air conditioners .
- **Vapor absorption refrigeration:** This method uses a vapor, such as ammonia or water, as the refrigerant, and a liquid, such as water or lithium bromide, as the absorbent. Vapor absorption refrigeration works by absorbing the vapor into the liquid, creating a solution, and then separating the vapor from the liquid by heating the solution, and repeating the cycle. Vapor absorption refrigeration is similar to vapor compression refrigeration, but it uses heat instead of mechanical work to drive the refrigeration process. Vapor absorption refrigeration is used for solar refrigeration, waste heat recovery, and refrigeration in remote areas .
- **Thermoelectric refrigeration:** This method uses

Construction and Working of Domestic Refrigerator

- A domestic refrigerator is a device that uses the vapor compression cycle to maintain a low temperature inside a closed cabinet for storing food and other items.
- The main components of a domestic refrigerator are:
 - A hermetically sealed compressor that compresses the refrigerant gas and increases its pressure and temperature.
 - A condenser that is a coil of metal pipe located at the back of the refrigerator. It cools down the hot refrigerant gas and condenses it into a liquid.
 - A capillary tube that is a narrow tube that connects the condenser and the evaporator. It reduces the pressure and temperature of the liquid refrigerant by throttling it.
 - An evaporator that is another coil of metal pipe located inside the freezer box. It absorbs heat from the contents of the refrigerator and evaporates the liquid refrigerant into a gas.
 - A thermostat that is a device that senses the temperature inside the refrigerator and controls the operation of the compressor.
 - An accumulator that is a vessel that collects any liquid refrigerant that may have escaped from the evaporator.
 - A drier and strainer that is a device that removes any moisture and impurities from the refrigerant.
- The working principle of a domestic refrigerator is as follows:
 - The compressor draws the low-pressure and low-temperature refrigerant gas from the evaporator and compresses it to a high-pressure and high-temperature gas.
 - The hot refrigerant gas flows through the condenser, where it releases its heat to the surrounding air and becomes a high-pressure and moderate-temperature liquid.
 - The liquid refrigerant passes through the capillary tube, where it experiences a pressure drop and a temperature drop. It becomes a low-pressure and low-temperature liquid.
 - The cold liquid refrigerant enters the evaporator, where it absorbs heat from the items inside the refrigerator and changes into a low-pressure and low-temperature gas.
 - The cycle repeats until the desired temperature is reached inside the refrigerator. The thermostat switches off the compressor when the temperature is low enough and switches it on again when the temperature rises.

Concept of Heat Pump

- A heat pump is a device that can provide heat to a building by transferring thermal energy from the outside using a refrigeration cycle.
- A heat pump is basically a heat engine run in the reverse direction. In other words, a heat pump is a device that is used to transfer heat energy to a thermal reservoir.
- A heat pump differs from other HVAC systems because it uses energy to pull heat from the outside and transfers it to the inside. It goes through a process of compression and exchange to increase the air's temperature and reverses the process to decrease the air's temperature.
- A heat pump is part of a home heating and cooling system and is installed outside your home. Like an air conditioner such as central air, it can cool your home, but it's also capable of providing heat.
- The concept behind heat pumps is simple: powered by electricity, they move heat around to either cool or heat buildings. It's not a new idea—they were invented in the 1850s and have been used in homes since the 1960s.
- A heat pump can be classified into different types based on the source and sink of heat, such as air-to-air, air-to-water, water-to-air, water-to-water, ground-to-air, and ground-to-water.
- A heat pump can be used for various purposes, such as space heating, space cooling, water heating, dehumidification, and refrigeration.
- A heat pump has many advantages, such as high efficiency, low operating cost, environmental friendliness, and versatility. However, it also has some disadvantages, such as high initial cost, noise, maintenance, and dependence on external factors.

Air-Conditioning

- Air-conditioning is the process of treating air so as to control simultaneously its temperature, humidity, purity and distribution to meet the requirements of the conditioned space, such as comfort and health of human beings and other needs of the situation.
- Air-conditioning can also be defined as the science which deals with supply and maintaining desirable internal atmospheric condition irrespective of external condition.
- Air-conditioning is a member of a family of systems and techniques that provide heating, ventilation, and air conditioning (HVAC).
- The principle of air-conditioning is based on the laws of thermodynamics. An air-conditioner operates using the refrigeration cycle. Specific refrigerants are needed as the working fluid in the refrigeration cycle. An air-conditioner goes through 4 processes; compression, condensation, expansion, and evaporation .
- The basic working principle of an air-conditioner is as follows:

- The refrigerant enters the compressor as a low-pressure vapor and is compressed to a high-pressure vapor.
- The high-pressure vapor then enters the condenser, where it releases heat to the surrounding air and condenses to a high-pressure liquid.
- The high-pressure liquid then passes through an expansion valve, where it reduces its pressure and temperature and becomes a low-pressure liquid.
- The low-pressure liquid then enters the evaporator, where it absorbs heat from the indoor air and evaporates to a low-pressure vapor.
- The low-pressure vapor then returns to the compressor and the cycle repeats.
- The main components of an air-conditioner are:
 - Compressor: It is the heart of the system that pumps the refrigerant through the cycle.
 - Condenser: It is the heat exchanger that rejects the heat from the refrigerant to the outside air.
 - Expansion valve: It is the device that controls the flow and pressure of the refrigerant entering the evaporator.
 - Evaporator: It is the heat exchanger that absorbs the heat from the indoor air and cools it down.
 - Refrigerant: It is the working fluid that circulates in the cycle and changes its phase from vapor to liquid and vice versa.
- The types of air-conditioning systems are :
 - Window air-conditioner: It is a self-contained unit that is installed in a window or a wall and cools a single room or a small area.
 - Split air-conditioner: It consists of two units, an indoor unit and an outdoor unit, that are connected by pipes. The indoor unit contains the evaporator and the expansion valve, while the outdoor unit contains the compressor and the condenser. It can cool one or more rooms or a large area.
 - Central air-conditioner: It is a system that cools the entire building or a complex of buildings using ducts and vents. It has a large compressor and condenser unit that is located outside the building, and a network of evaporator coils and expansion valves that are located inside the building. It can provide uniform temperature and humidity control throughout the building.
 - Packaged air-conditioner: It is a system that combines the compressor, condenser, evaporator, and expansion valve in a single unit that is installed on the roof or the ground. It can cool a large area or multiple rooms using ducts and vents.
 - Variable refrigerant flow (VRF) air-conditioner: It is a system that uses a single outdoor unit that contains multiple compressors and condensers, and multiple indoor units that contain evaporators and expansion valves. It can vary the refrigerant flow and temperature to each indoor unit according to the cooling demand. It can provide individual temperature and humidity control for each room or zone.

- Chilled water air-conditioner: It is a system that uses water as the secondary refrigerant that is chilled by the primary refrigerant in a chiller unit. The chilled water is circulated to the air handling units (AHUs) or fan coil units (FCUs) that are located in the conditioned space. The AHUs or FCUs contain coils that cool the air by exchanging heat with the chilled water. It can cool a large area or multiple rooms using ducts and vents.
- Vapor absorption air-conditioner: It is a system that uses heat as the driving force instead of electricity. It uses a solution of refrigerant and absorbent (such as water and lithium bromide) that undergoes absorption and desorption processes in the generator and the absorber

Its meaning and application

- Refrigeration is a process where thermal energy is transferred from a place with lower temperature to a place with higher temperature using energy, against the natural flow of heat .
- Air conditioning is a type of refrigeration which is used to cool large volumes inhabited by people. It also involves controlling the humidity, ventilation, and air quality of the space .
- The main purpose of using refrigeration and air conditioning is to lower the temperature of a particular area compared to the surroundings. The application of these can be seen in various domestic and industrial appliances such as ACs and refrigerators.
- Refrigeration and air conditioning are critical to the health, comfort, and welfare of humanity. They are used for preserving food, medicine, and other perishable goods, as well as for providing thermal comfort and indoor air quality in buildings, vehicles, and public spaces.
- Refrigeration and air conditioning systems are based on the principles of thermodynamics, heat transfer, and fluid mechanics. They consist of four main components: a compressor, a condenser, an expansion device, and an evaporator.
- Refrigeration and air conditioning systems can be classified into different types according to the refrigerant used, the cycle configuration, the heat rejection method, the cooling capacity, and the application area. Some examples of refrigeration and air conditioning systems are vapor compression systems, vapor absorption systems, gas cycle systems, air cycle systems, and thermoelectric systems.

Humidity

Humidity is a measure of the amount of water vapor present in the air. Water vapor, the gaseous state of water, is generally invisible to the human eye.

Humidity indicates the likelihood for precipitation, dew, or fog to be present. Humidity depends on the temperature and pressure of the system of interest.

There are three primary measurements of humidity that are widely employed: absolute, relative, and specific.

- **Absolute humidity** describes the water content present in the air and is expressed in either gram per cubic meter or grams per kilogram. The absolute humidity in the atmosphere ranges from near zero to roughly 30 grams per cubic metre. Absolute humidity is not affected by the temperature or pressure of the air, but it changes when the air expands or contracts.
- **Relative humidity** is the ratio of the partial pressure of water vapor to the equilibrium vapor pressure of water at the same temperature. It is expressed as a percentage and indicates how close the air is to saturation. Relative humidity depends on both the temperature and the pressure of the air, and it changes when the air is heated or cooled. Relative humidity is often used to describe the comfort level of the air, as high humidity can make people feel hot and sticky, while low humidity can cause dryness and irritation.
- **Specific humidity** is the ratio of water vapor mass to total moist air mass. It is expressed in grams of water vapor per kilogram of moist air. Specific humidity is a measure of the actual amount of water vapor in the air, regardless of the air's temperature or pressure. Specific humidity is conserved in an air parcel as long as no water is added or removed. Specific humidity is useful for comparing the humidity of different air masses at different locations and altitudes.

Humidity can be measured by various instruments, such as hygrometers, psychrometers, dew point meters, and humidity sensors. These instruments can measure one or more of the humidity parameters mentioned above, depending on their design and principle of operation. Humidity measurement is important for various applications, such as weather forecasting, air conditioning, industrial processes, agriculture, and health.

Dry Bulb Temperature

- Dry bulb temperature (DBT) is the temperature of air measured by a thermometer freely exposed to the air, but shielded from radiation and moisture.
- DBT is the temperature that is usually thought of as air temperature, and it is the true thermodynamic temperature.
- DBT is directly proportional to the mean kinetic energy of the air molecules.
- DBT is an indicator of heat content and is shown along the bottom axis of the psychrometric chart or along the left side of the Mollier diagram.

- DBT is also called “dry bulb” because the air temperature is indicated by a thermometer not affected by the moisture of the air.
- DBT is contrasted with wet bulb temperature and dew point temperature, which are related to the humidity of the air.

Wet Bulb Temperature

- Wet bulb temperature is the temperature read by a thermometer covered in water-soaked cloth over which air is passed.
- Wet bulb temperature is the lowest temperature that can be reached by the evaporation of water into the air at a constant pressure.
- Wet bulb temperature is the temperature of a parcel of air cooled to saturation by the evaporation of water into it, with the latent heat supplied by the parcel.
- Wet bulb temperature is the temperature of an object that can be achieved through evaporative cooling, assuming good air flow and that the ambient air temperature remains the same.
- Wet bulb temperature is an important parameter in psychrometrics, the study of the thermodynamic properties of moist air and its effect on materials and human comfort.
- Wet bulb temperature can be measured by using a wet-bulb thermometer, a sling psychrometer, or a hygrometer .
- Wet bulb temperature can be used to calculate the relative humidity, dew point temperature, humid heat, and enthalpy of moist air .
- Wet bulb temperature can also be used to estimate the heat stress on humans and animals, as it reflects the ability of the body to cool itself by sweating.
- Wet bulb temperature is lower than the dry bulb temperature (the ambient temperature) unless the air is fully saturated (100% relative humidity), in which case they are equal .
- Wet bulb temperature is always lower than the dew point temperature, which is the temperature at which the air becomes saturated by cooling at a constant pressure .

Dew point temperatures

- Dew point temperature is the temperature at which the air becomes saturated with water vapor, assuming constant air pressure and water content .
- When the air is cooled below the dew point temperature, the excess water vapor will condense into liquid water, forming dew, fog, or clouds .
- Dew point temperature depends on the amount of water vapor in the air, or the relative humidity. The higher the relative humidity, the higher the dew point temperature, and vice versa .

- Dew point temperature is an indicator of the comfort level and the quality of the air. High dew point temperatures make the air feel humid and sticky, while low dew point temperatures make the air feel dry and comfortable .
- Dew point temperature is also important for refrigeration and air-conditioning systems, as it determines the amount of moisture that can be removed from the air by cooling it. The lower the dew point temperature, the less moisture the air can hold, and the more efficient the cooling process .

Comfort Conditions

- Comfort conditions are the conditions of the indoor environment that are conducive to human health and well-being .
- Comfort conditions depend on various factors, such as temperature, humidity, air movement, air purity, and personal preferences .
- Comfort conditions are defined by the ASHRAE Standard 55, which specifies the acceptable ranges of these factors for different types of spaces and activities.
- Comfort conditions can be achieved by using refrigeration and air conditioning systems, which can control the temperature, humidity, and quality of the air supplied to the indoor spaces .
- Refrigeration and air conditioning systems can be classified into two types: comfort cooling and precision cooling .
- Comfort cooling systems are designed to provide comfortable conditions for human occupants, and have a lower and more variable heat load and a higher need for dehumidification .
- Precision cooling systems are designed to provide precise and stable conditions for sensitive equipment or processes, and have a higher and more constant heat load and a lower need for dehumidification .
- Comfort cooling systems can be further divided into two types: central systems and unitary systems.
- Central systems use a single or multiple refrigeration units to produce chilled water or air, which is then distributed to various zones or rooms through pipes or ducts.
- Unitary systems use individual refrigeration units to cool and condition the air for each zone or room, and do not require any distribution network.
- Some examples of comfort cooling systems are window air conditioners, split air conditioners, packaged air conditioners, fan coil units, and air handling units.

Construction and Working of Window Air Conditioner

- A window air conditioner is a small unit that can be mounted in a window or through a wall to cool a single room or space.
- It consists of the following main components :
 - **Compressor:** A device that compresses the refrigerant (usually R-22 or R-410A) from low pressure and temperature to high pressure and temperature.
 - **Condenser:** A coil of copper tubing that transfers heat from the hot refrigerant to the outside air. A fan blows air over the condenser coil to help the heat transfer.
 - **Expansion device:** A device that reduces the pressure and temperature of the refrigerant before it enters the evaporator. It can be a capillary tube or a thermostatic expansion valve (TXV).
 - **Evaporator:** A coil of copper tubing that absorbs heat from the room air and cools it. A fan blows air over the evaporator coil to help the heat transfer.
 - **Filter:** A device that removes dust, dirt, and other impurities from the room air before it passes through the evaporator.
 - **Drier:** A device that removes moisture from the refrigerant to prevent corrosion and freezing.
- The working principle of a window air conditioner is based on the vapour compression cycle :
 - The compressor pumps the refrigerant from the evaporator to the condenser, increasing its pressure and temperature.
 - The refrigerant releases heat to the outside air as it flows through the condenser coil, changing from vapour to liquid.
 - The liquid refrigerant passes through the expansion device, which reduces its pressure and temperature, making it ready to absorb heat from the room air.
 - The refrigerant absorbs heat from the room air as it flows through the evaporator coil, changing from liquid to vapour.
 - The vapour refrigerant returns to the compressor, completing the cycle.
- The advantages of a window air conditioner are:
 - It is easy to install and remove, requiring only a standard electrical outlet and a suitable window or wall opening.
 - It is relatively inexpensive and energy-efficient, compared to other types of air conditioners.
 - It does not occupy floor space or require ductwork, making it suitable for small rooms or spaces.

Unit 4 - Introduction to Fluid Mechanics and Applications

- Fluid mechanics is the branch of physics concerned with the mechanics of fluids (liquids, gases, and plasmas) and the forces on them.
- Fluid mechanics has a wide range of applications in engineering, biological systems, and astrophysics .
- Some examples of fluid mechanics applications are:
 - Hydraulic and aeronautical engineering: designing pumps, turbines, pipes, valves, aircrafts, rockets, etc. that involve fluid flow and pressure.
 - Chemical engineering: designing reactors, separators, mixers, etc. that involve fluid transport and reaction.
 - Meteorology: predicting weather patterns, climate change, etc. that involve fluid dynamics and thermodynamics.
 - Zoology: studying the locomotion, circulation, respiration, etc. of animals that involve fluid mechanics and biomechanics.
- Fluid mechanics can be divided into two main branches: fluid statics and fluid dynamics.
 - Fluid statics deals with fluids at rest and the forces exerted by them on solid surfaces or other fluids.
 - Fluid dynamics deals with fluids in motion and the forces that cause and result from their motion.
- Fluid mechanics can also be classified based on the type of fluid, the flow regime, and the flow geometry.
 - The type of fluid can be Newtonian or non-Newtonian, depending on whether the fluid viscosity is constant or varies with the shear stress.
 - The flow regime can be laminar or turbulent, depending on whether the fluid flow is smooth and orderly or chaotic and irregular.
 - The flow geometry can be one-dimensional, two-dimensional, or three-dimensional, depending on whether the fluid flow is confined to a single direction, a plane, or a volume.

Introduction for the notes of the Unit 4 - Introduction to Fluid Mechanics and Applications in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

- Fluid mechanics is the branch of science that deals with the behavior of fluids (liquids and gases) at rest and in motion.
- Fluid mechanics has a wide range of applications in engineering, such as aerodynamics, hydraulics, lubrication, blood flow, weather prediction, etc.
- Fluid mechanics can be divided into two main subfields: fluid statics and

fluid dynamics.

- Fluid statics is the study of fluids at rest or in equilibrium. It involves the analysis of forces and pressures acting on fluid elements and bodies immersed in fluids.
- Fluid dynamics is the study of fluids in motion. It involves the analysis of velocity, acceleration, pressure, density, temperature, and other properties of fluid flows.
- Fluid dynamics can be further classified into two categories: incompressible and compressible flows.
- Incompressible flows are those in which the density of the fluid remains constant or changes negligibly. Examples are water flow, air flow at low speeds, etc.
- Compressible flows are those in which the density of the fluid changes significantly due to variations in pressure and temperature. Examples are air flow at high speeds, gas flow in pipes, etc.
- Some of the fundamental concepts and principles of fluid mechanics are:
 - Fluid properties: such as density, viscosity, surface tension, specific weight, specific gravity, etc.
 - Fluid kinematics: the description of fluid motion without considering the forces causing it. It involves the concepts of streamlines, pathlines, streaklines, velocity field, acceleration field, etc.
 - Fluid dynamics: the analysis of fluid motion by considering the forces causing it. It involves the concepts of pressure, shear stress, normal stress, momentum, energy, etc.
 - Continuity equation: the principle of conservation of mass for a fluid flow. It states that the rate of change of mass within a control volume is equal to the net rate of mass flow across its boundary.
 - Bernoulli's equation: the principle of conservation of energy for a steady, incompressible, inviscid, and irrotational flow along a streamline. It states that the sum of the pressure, kinetic energy, and potential energy per unit mass of the fluid is constant along the streamline.
 - Euler's equation: the equation of motion for an inviscid flow. It states that the rate of change of momentum of a fluid element is equal to the net force acting on it.
 - Navier-Stokes equation: the equation of motion for a viscous flow. It is a generalization of Euler's equation that includes the effects of viscosity and heat transfer.
 - Reynolds number: a dimensionless parameter that characterizes the type of flow. It is defined as the ratio of inertial forces to viscous forces in the flow. It indicates whether the flow is laminar (smooth and orderly) or turbulent (chaotic and irregular).
 - Boundary layer: the thin layer of fluid adjacent to a solid surface where the effects of viscosity are significant. It is characterized by a velocity gradient normal to the surface and a pressure gradient along the surface.
 - Drag: the force exerted by a fluid on a body moving through it in

- the direction opposite to the motion. It is caused by the pressure difference and the shear stress on the surface of the body.
- Lift: the force exerted by a fluid on a body moving through it in the direction perpendicular to the motion. It is caused by the pressure difference and the circulation around the body.

Fluid Properties for the Notes of the Unit 4 - Introduction to Fluid Mechanics and Applications in the Subject of Fundamentals of Mechanical Engineering

Fluid mechanics is the science concerned with the response of fluids to forces exerted upon them. Fluids are substances that can flow and deform under applied shear stress, such as liquids and gases. The study of fluid mechanics is important for many applications in engineering, such as hydraulic systems, aerodynamics, fluid power, and heat transfer.

The various properties of fluids that are relevant for fluid mechanics are :

- **Density or mass density:** The density or mass density of a fluid is defined as the ratio of the mass of a fluid to the volume of the fluid. It is denoted by the symbol ρ (rho) and has the units of kg/m^3 . Density is a measure of how much mass is contained in a given volume of fluid. Density varies with temperature and pressure for most fluids, except for incompressible fluids, such as water, which have a constant density.
- **Specific weight or weight density:** The specific weight or weight density of a fluid is defined as the ratio of the weight of a fluid to the volume of the fluid. It is denoted by the symbol γ (gamma) and has the units of N/m^3 . Specific weight is a measure of how much weight is exerted by a given volume of fluid due to gravity. Specific weight is related to density by the equation $\gamma = \rho g$, where g is the acceleration due to gravity.
- **Specific volume:** The specific volume of a fluid is defined as the ratio of the volume of a fluid to the mass of the fluid. It is denoted by the symbol v and has the units of m^3/kg . Specific volume is a measure of how much space is occupied by a given mass of fluid. Specific volume is the inverse of density, i.e., $v = 1/\rho$.
- **Specific gravity:** The specific gravity of a fluid is defined as the ratio of the density of a fluid to the density of a standard fluid, usually water at 4°C or air at standard atmospheric conditions. It is denoted by the symbol S or SG and has no units. Specific gravity is a measure of how heavy a fluid is compared to a reference fluid. Specific gravity is dimensionless and independent of the system of units.

- **Viscosity:** The viscosity of a fluid is defined as the measure of the resistance of a fluid to flow or deformation under applied shear stress. It is denoted by the symbol μ (mu) and has the units of Pa·s or N·s/m². Viscosity is a property that characterizes the internal friction of a fluid. Viscosity depends on the molecular structure and interactions of the fluid, as well as the temperature and pressure. Viscosity affects the flow behavior and energy losses of fluids in motion.

These are the main properties of fluids that are used in fluid mechanics. Other properties, such as surface tension, vapor pressure, compressibility, and thermal conductivity, may also be relevant for some applications. The knowledge of these properties is essential to make the study of fluid mechanics comprehensive and to solve practical problems involving fluids.

Pressure

- Pressure is the physical force exerted on an object per unit area .
- Pressure is denoted by the capital letter P or the lowercase letter p.
- The basic formula for pressure is $P = F / A$, where F is the force and A is the area .
- The SI unit of pressure is the Pascal (Pa), where $1 \text{ Pa} = 1 \text{ N/m}^2$, i.e. one Newton per square meter.
- Pressure can be increased by increasing the force or decreasing the area, and vice versa .
- Pressure can be measured using devices such as barometers, manometers, and pressure gauges .
- Pressure can be classified into different types, such as atmospheric pressure, gauge pressure, absolute pressure, hydrostatic pressure, and fluid pressure .
- Pressure is an important concept in fluid mechanics, as it affects the flow and behavior of fluids .

Density

- Density is the measurement of how tightly a material is packed together. It is defined as the mass per unit volume .
- The symbol for density is ρ (the Greek letter rho).
- The formula for density is $\rho = M / V$, where ρ is density, M is mass, and V is volume .
- Density is commonly expressed in units of grams per cubic centimetre (g/cm³) or kilograms per cubic metre (kg/m³). For example, the density of water is 1 g/cm³ or 1000 kg/m³, and Earth's density is 5.51 g/cm³ or 5510 kg/m³ .
- Density can also be expressed in other units, such as pounds per cubic foot (lb/ft³) or ounces per cubic inch (oz/in³), depending on the context

and the system of units used.

- Density is an intensive property, which means it does not depend on the size or shape of the object. For example, a gold coin and a gold bar have the same density, even though they have different masses and volumes.
- Density can be used to identify substances or to compare the compactness of different materials. For example, iron, platinum, and lead are examples of dense materials, while air, helium, and styrofoam are examples of less dense materials .
- Density can also be affected by temperature and pressure. For example, water expands when it freezes, so ice has a lower density than liquid water. Similarly, air becomes denser when it is compressed, so air at sea level has a higher density than air at high altitudes.

Dynamic and Kinematic Viscosity

- Viscosity is a property of fluids that measures their resistance to flow. It is also known as the internal friction of fluids.
- There are two types of viscosity: dynamic (or absolute) viscosity and kinematic viscosity.
- Dynamic viscosity evaluates the internal resistance of a fluid to flow when a certain force is applied. It is a measure of how much force is needed to make the fluid flow at a certain rate. It depends on the type of fluid and its temperature. The SI unit of dynamic viscosity is N s/m^2 or Pa s .
- Kinematic viscosity describes the ratio of dynamic viscosity to density. It is a measure of how fast the fluid moves when a certain force is applied. It incorporates the fluid's weight or density as part of its measurement. It also depends on the type of fluid and its temperature. The SI unit of kinematic viscosity is m^2/s .
- Two fluids with the same value of dynamic viscosity can have different values of kinematic viscosity based on their density and vice versa. For example, honey and water have similar dynamic viscosity at room temperature, but honey has a higher density and therefore a higher kinematic viscosity than water.
- Some examples of fluids with high dynamic viscosity are honey, molasses, glycerin, and motor oil. Some examples of fluids with low dynamic viscosity are water, air, gasoline, and alcohol.
- Some examples of fluids with high kinematic viscosity are honey, molasses, glycerin, and motor oil. Some examples of fluids with low kinematic viscosity are air, gasoline, alcohol, and water.
- The viscosity of fluids affects their flow behavior and their applications in

engineering and science. For example, fluids with high viscosity are used as lubricants, adhesives, and coatings, while fluids with low viscosity are used as fuels, solvents, and refrigerants.

Specific Gravity

- Specific gravity is a dimensionless quantity that compares the density of a substance with the density of a reference substance, usually water.
- Specific gravity can be calculated using the following formula :

$$SG = \frac{\rho_{\text{substance}}}{\rho_{\text{water}}}$$

– Where SG is the specific gravity, $\rho_{\text{substance}}$ is the density of the substance, and ρ_{water} is the density of water.

- The unit of specific gravity is the same as the unit of density, which is kg/m^3 or g/cm^3 .
- The specific gravity of water at 4°C is 1, because the density of water at that temperature is 1000 kg/m^3 or 1 g/cm^3 .
- The specific gravity of a substance can be used to determine if it will sink or float in water. A substance with a specific gravity less than 1 will float, while a substance with a specific gravity greater than 1 will sink.
- For example, the specific gravity of gold is 19.3, which means that gold is 19.3 times denser than water. Therefore, gold will sink in water.
- Some factors that affect the specific gravity of a substance are temperature, pressure, and purity. The specific gravity of a substance will change with the change in these factors.

Newtonian and Non-Newtonian Fluids

- Fluids can be divided into two major types: **Newtonian** and **non-Newtonian**.
- **Newtonian fluids** are the ones that obey **Newton's law of constant viscosity**. These fluids have a **constant viscosity** and **zero shear rate and shear stress**. Examples of Newtonian fluids are water, mineral oil, gasoline, and alcohol.
- **Non-Newtonian fluids** are those that do not have a constant viscosity and have a **variable relationship with shear stress**. These fluids can change their viscosity when under force to either more liquid or more solid. Examples of non-Newtonian fluids are ketchup, blood, yogurt, and toothpaste.

- The importance of knowing the type of fluid is that it affects the **processing** and **handling** of the fluid. For example, non-Newtonian fluids may require different types of pumps, pipes, valves, and mixers than Newtonian fluids. Non-Newtonian fluids may also exhibit different flow behaviors such as **thixotropy**, **rheopexy**, **dilatancy**, and **Bingham plasticity**.
- Thixotropy is the property of some non-Newtonian fluids to become less viscous over time when shaken, agitated, or otherwise stressed. Rheopexy is the opposite of thixotropy, where some non-Newtonian fluids become more viscous over time when subjected to stress. Dilatancy is the property of some non-Newtonian fluids to become more viscous when subjected to shear stress. Bingham plasticity is the property of some non-Newtonian fluids to behave like a solid until a certain amount of stress is applied, then flow like a liquid.
- To summarize, Newtonian fluids have a constant viscosity and zero shear rate and shear stress, while non-Newtonian fluids have a variable viscosity and a complex relationship with shear stress. Non-Newtonian fluids can also exhibit different flow behaviors depending on the type and amount of stress applied. Knowing the type of fluid is important for processing and handling purposes.

Pascal's Law and Continuity Equation

Pascal's Law

- Pascal's law states that a pressure applied to a fluid in a closed container is transmitted equally to every point of the fluid and the walls of the container.
- The mathematical expression of Pascal's law is:

$$P = \frac{F}{A}$$

Where P is the pressure, F is the force, and A is the area.

- Pascal's law can be used to explain how hydraulic systems work, such as hydraulic brakes, lifts, and presses.
- Pascal's law can also be applied to calculate the pressure at different depths in a fluid, such as water or air. The formula is:

$$\Delta P = \rho g \Delta h$$

Where ΔP is the change in pressure, ρ is the density of the fluid, g is the acceleration due to gravity, and Δh is the change in height.

Continuity Equation

- The continuity equation expresses the relationship between mass flow rates at different points in a fluid system under steady-state flow conditions.
- The continuity equation is based on the principle of conservation of mass, which states that the mass of a fluid cannot be created or destroyed in a closed system.
- The mathematical expression of the continuity equation is:

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

Where ρ is the density, A is the cross-sectional area, and v is the velocity of the fluid at points 1 and 2.

- The continuity equation can be used to analyze the flow of fluids through pipes, nozzles, orifices, and other devices.
- The continuity equation can also be written in terms of volume flow rate, which is the product of cross-sectional area and velocity. The formula is:

$$Q_1 = Q_2$$

Where Q is the volume flow rate at points 1 and 2.

Working principles of hydraulic turbines (Pelton Wheel and Francis) and pumps (Centrifugal and Reciprocating) and their classifications and hydraulic lift

Hydraulic Turbines

- A hydraulic turbine is a device that converts the kinetic and potential energy of water into mechanical work.
- The mechanical work can be used to drive an electric generator or other machinery.
- A hydraulic turbine consists of a runner (a rotating wheel with blades or buckets), a casing (a stationary housing that guides the water flow), and a draft tube (a pipe that connects the turbine exit to the tailrace).
- The working principle of a hydraulic turbine is based on Newton's second law of motion, which states that the force acting on a body is equal to the rate of change of its momentum.
- When water flows through the turbine, it exerts a force on the runner blades or buckets, causing them to rotate. The water also loses some

of its kinetic and potential energy, which is transferred to the runner as mechanical work.

- The amount of work done by the turbine depends on the mass flow rate, the head (the difference in elevation between the inlet and outlet of the turbine), and the efficiency of the turbine.
- There are two main types of hydraulic turbines: impulse turbines and reaction turbines.
- Impulse turbines operate under high head and low flow conditions. They use the kinetic energy of a jet of water to rotate the runner. The pressure of the water remains constant throughout the turbine. Examples of impulse turbines are Pelton wheel and Turgo wheel.
- Reaction turbines operate under low head and high flow conditions. They use both the kinetic and potential energy of the water to rotate the runner. The pressure of the water decreases as it passes through the turbine. Examples of reaction turbines are Francis turbine and Kaplan turbine.

Pelton Wheel

- A Pelton wheel is a type of impulse turbine that consists of a circular wheel with spoon-shaped buckets mounted on its rim.
- The water is supplied from a nozzle at high speed and strikes the buckets tangentially at one point, causing them to rotate.
- The water then deflects at an angle and leaves the buckets with reduced speed and energy.
- The Pelton wheel is suitable for high head and low flow applications, such as hydroelectric power plants in mountainous regions.
- The advantages of a Pelton wheel are high efficiency, simple design, and easy maintenance.
- The disadvantages of a Pelton wheel are low specific speed, large size, and sensitivity to dirt and debris in the water.

Francis Turbine

- A Francis turbine is a type of reaction turbine that consists of a spiral-shaped casing, a runner with radial curved blades, and a draft tube.
- The water enters the casing at the outer edge of the runner and flows radially inward, following the shape of the blades.
- The water then turns 90 degrees and exits the runner axially through the draft tube.
- The Francis turbine is suitable for medium head and medium flow applications, such as irrigation, water supply, and industrial processes.
- The advantages of a Francis turbine are high efficiency, wide range of operation, and compact size.
- The disadvantages of a Francis turbine are complex design, high cost, and cavitation risk.

Hydraulic Pumps

- A hydraulic pump is a device that converts the mechanical energy from an engine or motor into hydraulic energy (pressure and flow) in a fluid.
- The hydraulic energy can be used to power hydraulic cylinders, motors, valves, and other components in a hydraulic system.
- A hydraulic pump consists of a driving mechanism (such as a crankshaft, cam, or piston), a pumping element (such as a gear, vane, or piston), and a suction and discharge port.
- The working principle of a hydraulic pump is based on the principle of displacement. For each rotating or reciprocating cycle of the pumping element, a fixed volume of fluid is displaced or delivered from the suction port to the discharge port.
- The pressure of the fluid is determined by the resistance of the hydraulic system, while the flow rate of the fluid is determined by the speed and size of the pumping element.
- There are two main types of hydraulic pumps: positive displacement pumps and non-positive displacement pumps.
- Positive displacement pumps deliver a constant flow of fluid for each cycle of the pumping element, regardless of the pressure. They are used for high pressure and low flow applications, such as hydraulic presses, jacks, and brakes.
- Non-positive displacement pumps deliver a variable flow of fluid for each cycle of the pumping element, depending on the pressure. They are used for low pressure and high flow applications, such as cooling, lubrication, and filtration.

Unit 5 - Introduction to Measurement and Mechatronics

- Measurement is the process of obtaining quantitative information about a physical phenomenon or property using a device or instrument.
- Mechatronics is the interdisciplinary field that integrates mechanical, electrical, and computer engineering to design and control systems that involve sensors, actuators, and feedback.
- The basic elements of a mechatronic system are:
 - Input: the physical quantity or signal that is measured or controlled by the system.
 - Sensor: the device that converts the input into an electrical signal that can be processed by the system.
 - Signal conditioning: the circuitry that modifies, filters, amplifies, or converts the sensor signal to make it suitable for the system.
 - Processor: the device that performs computations, logic, or decision making based on the sensor signal and the desired output.
 - Actuator: the device that converts the electrical signal from the processor into a physical output that affects the system or the environment.

- Output: the physical quantity or signal that is the result of the system’s action or control.
- Feedback: the process of comparing the output with the desired or reference value and adjusting the input accordingly to reduce the error or deviation.
- The advantages of mechatronic systems are:
 - Higher performance, efficiency, accuracy, and reliability.
 - Greater flexibility, adaptability, and functionality.
 - Reduced size, weight, and cost.
 - Easier maintenance and troubleshooting.
- The challenges of mechatronic systems are:
 - Complexity and interdependence of components and subsystems.
 - Need for multidisciplinary knowledge and skills.
 - Trade-offs and compromises among conflicting design criteria and constraints.
 - Uncertainty and variability of inputs, outputs, and disturbances.
 - Nonlinearities and dynamics of the system behavior and response.

Introduction to Measurement

- Measurement is the process of **comparing** an unknown quantity with a known quantity .
- The result of a measurement is a **numeric value** with certain **units** .
- Measurement is **fundamental** to the sciences, engineering, construction, and other technical fields, as well as to almost all everyday activities.
- Measurement involves a **definition** of the quantity that is to be measured, and a **comparison** with some known quantity of the same kind.
- If the object or quantity to be measured is not accessible for direct comparison, it is **converted** or “transduced” into an analogous measurement signal.
- Measurement can be classified into different types, such as **direct** and **indirect**, **absolute** and **relative**, **static** and **dynamic**, etc.
- Measurement can also be categorized by the **physical quantity** that is being measured, such as **length**, **mass**, **capacity** (volume), **temperature**, **force**, **pressure**, **speed**, **electricity**, etc .
- Measurement requires the use of **instruments** or **devices** that can produce reliable and accurate results, such as **rulers**, **scales**, **thermometers**, **gauges**, **meters**, **sensors**, etc .
- Measurement also involves the use of **standards** or **units** that are universally accepted and consistent, such as the **International System of Units (SI)**, which is based on seven base units: **meter** (m) for length, **kilogram** (kg) for mass, **second** (s) for time, **ampere** (A) for electric current, **kelvin** (K) for temperature, **mole** (mol) for amount of substance, and **candela** (cd) for luminous intensity .
- Measurement also requires the use of **conversion** or **transformation**

methods that can change the units or scales of a measurement, such as **multiplication** or **division** by a factor, **addition** or **subtraction** of an offset, **exponentiation** or **logarithm**, **trigonometry**, etc .

- Measurement also involves the **analysis** and **interpretation** of the measurement results, such as **error** or **uncertainty** estimation, **statistical** or **graphical** representation, **comparison** or **correlation** with other measurements, **inference** or **prediction** of the underlying phenomena, etc.

Concept of Measurement

- Measurement is the process of comparing an unknown quantity with a known standard or unit.
- Measurement is essential for verifying the dimensions, properties, and performance of a product or a system.
- Measurement is also important for controlling and improving the quality and efficiency of a process or a system.
- The science of measurement is called metrology, which deals with the principles, methods, and standards of measurement.
- Measurement can be classified into two types: direct and indirect measurement.
 - Direct measurement is the process of obtaining the value of a quantity by comparing it with a standard unit directly, such as measuring the length of a rod with a ruler.
 - Indirect measurement is the process of obtaining the value of a quantity by using a mathematical relation with another quantity that can be measured directly, such as measuring the speed of a car by measuring the time and distance traveled.
- Measurement can also be classified into three categories: nominal, ordinal, and interval measurement.
 - Nominal measurement is the process of assigning names or labels to different categories of a quantity, such as measuring the gender of a person by using male or female labels.
 - Ordinal measurement is the process of ranking the categories of a quantity in a certain order, such as measuring the hardness of a material by using a scale from soft to hard.
 - Interval measurement is the process of assigning numerical values to the categories of a quantity based on a fixed unit and a zero point, such as measuring the temperature of a substance by using a thermometer with degrees Celsius or Fahrenheit.
- Measurement involves four basic elements: the measurand, the instrument, the method, and the result.
 - The measurand is the quantity that is to be measured, such as the length of a rod, the mass of a ball, or the temperature of a substance.
 - The instrument is the device that is used to measure the measurand, such as a ruler, a balance, or a thermometer.

- The method is the procedure that is followed to measure the measurand using the instrument, such as placing the ruler along the rod, placing the ball on the balance, or inserting the thermometer into the substance.
- The result is the value that is obtained from the measurement, such as 10 cm, 50 g, or 25 °C.
- Measurement is subject to errors and uncertainties, which affect the accuracy and precision of the result.
 - Accuracy is the degree of closeness of the result to the true value of the measurand, such as measuring the length of a rod as 10.01 cm when the true value is 10.00 cm.
 - Precision is the degree of consistency or repeatability of the result, such as measuring the length of a rod as 10.01 cm, 10.02 cm, and 10.03 cm in three trials.
 - Errors are the deviations of the result from the true value of the measurand, which can be caused by various factors, such as the instrument, the method, the environment, or the operator.
 - Uncertainties are the estimates of the possible errors in the result, which can be expressed by a range or an interval, such as 10.01 ± 0.01 cm.

Error in measurements

- Error in measurements is the deviation of the measured value from the true value of the quantity.
- Error can be classified into two types: systematic error and random error.
- Systematic error is the error that is constant and predictable in a measurement. It can be caused by faulty instruments, incorrect calibration, improper methods, or human errors. Systematic error can be reduced by correcting the source of error or applying a suitable correction factor.
- Random error is the error that is unpredictable and varies in magnitude and direction in a measurement. It can be caused by environmental factors, fluctuations in the instrument, or human errors. Random error can be reduced by increasing the number of measurements and taking the average value.
- The accuracy of a measurement is the degree of closeness of the measured value to the true value. It is affected by both systematic and random errors. A measurement can be accurate but not precise, or precise but not accurate, or both accurate and precise, or neither accurate nor precise.
- The precision of a measurement is the degree of repeatability of the measured value. It is affected by random errors only. A measurement can be precise if the measured values are close to each other, even if they are far from the true value.
- The absolute error of a measurement is the difference between the measured value and the true value. It can be positive or negative, depending

on whether the measured value is greater or less than the true value.

- The relative error of a measurement is the ratio of the absolute error to the true value. It is usually expressed as a percentage. It indicates the magnitude of the error relative to the true value.
- The standard deviation of a measurement is a measure of the dispersion of the measured values around the mean value. It is calculated by taking the square root of the variance, which is the average of the squared differences of the measured values from the mean value. It indicates the uncertainty of the measurement. A smaller standard deviation means a more precise measurement.

Calibration

- Calibration is the action or process of comparing an instrument or experimental readings with a standard of known accuracy and adjusting the instrument or readings accordingly .
- Calibration ensures that the instrument or readings are accurate, reliable, and consistent with the intended measurements .
- Calibration is usually performed by using a calibration standard, which is a device or material that has a known or assigned value of a physical quantity, such as length, mass, temperature, or voltage .
- The calibration standard is normally traceable to a national or international standard held by a metrology body, such as the National Institute of Standards and Technology (NIST) or the International Bureau of Weights and Measures (BIPM) .
- The calibration process involves two steps:
 - In the first step, a relation is established between the quantity values with measurement uncertainties provided by the calibration standard and the corresponding indications with associated measurement uncertainties of the instrument or readings .
 - In the second step, this relation is used to obtain a measurement result from an indication, or to adjust the instrument or readings to reduce the measurement error .
- Calibration can be performed in different ways, depending on the type and purpose of the instrument or readings. Some common methods of calibration are:
 - Direct comparison: The instrument or readings are compared with the calibration standard directly, without any intermediate device or calculation.
 - Indirect comparison: The instrument or readings are compared with the calibration standard indirectly, using an intermediate device or calculation, such as a transfer standard, a bridge circuit, or a calibration curve.
 - Self-calibration: The instrument or readings are calibrated by using a built-in or external feature that allows the instrument or readings to

- adjust themselves automatically or manually, such as a zero button, a span button, or a calibration mode.
- Software calibration: The instrument or readings are calibrated by using a software program that corrects the instrument or readings based on a mathematical model or algorithm, such as a linearization, a compensation, or a correction factor.
- Calibration is an essential part of measurement and mechatronics, as it ensures the quality and accuracy of the data and information obtained from the instruments or readings, and enables the proper functioning and performance of the mechatronic systems .

Measurements of Pressure (Bourdon Tube Pressure and U-Tube Manometer)

- Pressure is the force per unit area exerted by a fluid on a surface.
- Pressure can be measured by various devices that use the principle of balancing the fluid force with a known force.
- Two common devices for measuring pressure are the Bourdon tube pressure gauge and the U-tube manometer.

Bourdon Tube Pressure Gauge

- A Bourdon tube pressure gauge consists of a hollow metal tube that is bent into a circular or elliptical shape and closed at one end.
- The other end of the tube is connected to a pointer that moves over a calibrated scale.
- When the tube is exposed to a fluid pressure, it tends to straighten out due to the internal force of the fluid.
- The amount of straightening is proportional to the fluid pressure and is indicated by the pointer on the scale.
- The Bourdon tube pressure gauge can measure both gauge and absolute pressure, depending on the reference pressure at the closed end of the tube.
- The Bourdon tube pressure gauge can measure pressure up to 1600 bar and is compatible with almost any fluid medium.

U-Tube Manometer

- A U-tube manometer consists of a U-shaped tube filled with a liquid of known density, such as mercury or water.
- One end of the tube is connected to the fluid whose pressure is to be measured, and the other end is open to the atmosphere or another reference pressure.
- The difference in height of the liquid columns in the two arms of the tube is proportional to the pressure difference between the two ends of the tube.

- The U-tube manometer can measure both gauge and differential pressure, depending on the reference pressure at the open end of the tube.
- The U-tube manometer can measure pressure up to a few bar and is suitable for low-pressure and low-viscosity fluids .

Temperature (Thermocouple and Optical Pyrometer)

- Temperature is a measure of the average kinetic energy of the molecules in a substance.
- Temperature measurement is important for many industrial processes, such as metallurgy, chemical engineering, and manufacturing.
- There are different methods of measuring temperature, depending on the range, accuracy, and application required.
- Two common methods of temperature measurement are thermocouple and optical pyrometer.

Thermocouple

- A thermocouple is a device that consists of two dissimilar metals joined at one end, forming a junction.
- When the junction is heated or cooled, a voltage is generated across the open ends of the wires, proportional to the temperature difference between the junction and the reference point.
- The voltage can be measured by a voltmeter or a thermometer, and calibrated to give the temperature of the junction.
- Thermocouples are widely used for measuring temperatures from -200°C to 2300°C, with high accuracy and fast response time.
- Thermocouples are also relatively cheap, simple, and durable, and can be used in harsh environments.
- Some disadvantages of thermocouples are that they require a reference point, which may drift or be affected by ambient temperature, and that they are susceptible to electromagnetic interference and corrosion.

Optical Pyrometer

- An optical pyrometer is a device that measures the temperature of a hot object by comparing the intensity or color of the radiation emitted by the object with a known source of radiation, such as a lamp or a blackbody.
- The optical pyrometer has a lens, a filter, and a detector, which can be a human eye, a photodiode, or a camera.
- The optical pyrometer is aimed at the target object, and the filter is adjusted until the brightness or color of the target matches the source.
- The temperature of the target can then be read from a scale or a display, based on the filter position or the wavelength of the radiation.

- Optical pyrometers are useful for measuring temperatures above 700°C, where contact methods are impractical or inaccurate.
- Optical pyrometers are also non-invasive, fast, and easy to use, and can measure the temperature of moving or inaccessible objects.
- Some disadvantages of optical pyrometers are that they require a clear line of sight to the target, which may be obstructed by dust, smoke, or steam, and that they are sensitive to the emissivity of the target, which may vary with the material, surface condition, and angle of view.

Mass Flow Rate (Venturi Meter and Orifice Meter)

- Mass flow rate is the amount of mass of a fluid passing through a cross-sectional area per unit time. It is usually denoted by $\dot{m}$ and has the units of kg/s or lbm/s.
- Venturi meter and orifice meter are devices that are used to measure the mass flow rate of a fluid by creating a pressure difference across a constriction in the flow path.
- Both devices are based on the application of the continuity equation and the Bernoulli's equation, which relate the pressure, velocity, and area of a fluid at different points along a streamline.
- The continuity equation states that the mass flow rate is constant for an incompressible fluid, so the product of the area and the velocity at any point is the same. This means that the velocity increases as the area decreases, and vice versa.
- The Bernoulli's equation states that the sum of the pressure, the kinetic energy, and the potential energy of a fluid is constant along a streamline, neglecting friction and heat transfer. This means that the pressure decreases as the velocity increases, and vice versa.
- Venturi meter is a device that consists of a converging section, a throat, and a diverging section. The fluid velocity is highest and the pressure is lowest at the throat, where the area is minimum. The pressure difference between the inlet and the throat is measured by a manometer or a differential pressure gauge, and is used to calculate the mass flow rate using the following formula:

$$\dot{m} = A_2 * \sqrt{2 * \rho * (P_1 - P_2) / (1 - (A_2 / A_1)^2)}$$

where A_1 and A_2 are the areas of the inlet and the throat, ρ is the density of the fluid, and P_1 and P_2 are the pressures at the inlet and the throat.

- Orifice meter is a device that consists of a thin plate with a hole in the

center, called an orifice, that is inserted in the flow path. The fluid velocity is highest and the pressure is lowest at the orifice, where the area is minimum. The pressure difference between the upstream and the downstream of the orifice is measured by a manometer or a differential pressure gauge, and is used to calculate the mass flow rate using the following formula:

$$\dot{m} = C_d * A * \sqrt{2 * \rho * (P_1 - P_2)}$$

where C_d is the discharge coefficient, which accounts for the losses due to friction and turbulence, A is the area of the orifice, ρ is the density of the fluid, and P_1 and P_2 are the pressures at the upstream and the downstream of the orifice.

- Venturi meter is more accurate than orifice meter because it has a lower discharge coefficient, which means less losses due to friction and turbulence. The discharge coefficient of a venturi meter is typically 0.96-0.99, while the discharge coefficient of an orifice meter is typically 0.6-0.62. Venturi meter also has a lower pressure drop, which means less energy loss and less pumping power required. The pressure drop of a venturi meter is about 10-20% of the pressure difference, while the pressure drop of an orifice meter is about 60-70% of the pressure difference.
- Venturi meter and orifice meter are suitable for measuring the mass flow rate of clean and dirty liquids, gases, and some slurries. They are simple, cheap, and easy to install and maintain. However, they have some limitations, such as sensitivity to the Reynolds number, the viscosity, and the temperature of the fluid, and the need for calibration and correction factors. Some alternatives to venturi meter and orifice meter are nozzle meter, pitot tube, rotameter, turbine meter, magnetic meter, ultrasonic meter, and Coriolis meter, which have different advantages and disadvantages depending on the application and the fluid properties.

Strain (Bonded and Unbonded Strain Gauge)

- Strain is the measure of deformation or change in dimensions of a body due to an applied force or stress.
- Strain can be expressed as the ratio of change in length to the original length of the body.
- Strain can be positive (tensile) or negative (compressive) depending on the direction of the applied force or stress.
- Strain can be measured by using a device called a strain gauge, which is an example of a passive transducer that converts a mechanical displacement into a change of resistance.
- A strain gauge consists of an insulating flexible backing which supports a metallic foil pattern. The foil pattern has a grid of parallel wires that are electrically continuous. The backing is attached to the surface of the body whose strain is to be measured by a suitable adhesive.

- When the body is subjected to strain, the backing and the foil pattern deform along with it, causing a change in the length and cross-sectional area of the wires. This results in a change in the electrical resistance of the strain gauge, which can be measured by a Wheatstone bridge circuit or a similar device.
- The change in resistance is proportional to the applied strain, and the proportionality constant is called the gauge factor. The gauge factor depends on the material and geometry of the strain gauge, and is usually provided by the manufacturer.
- There are two types of strain gauges: bonded and unbonded. Bonded strain gauges are directly attached to the surface of the body whose strain is to be measured, while unbonded strain gauges are not in direct contact with the body, but are suspended on a frame or carrier that is attached to the body.
- Bonded strain gauges have the advantage of being more sensitive, accurate, and stable than unbonded strain gauges, but they are also more prone to errors due to temperature changes, creep, hysteresis, and fatigue. Unbonded strain gauges have the advantage of being less affected by these factors, but they are also less sensitive, accurate, and stable than bonded strain gauges, and they require a more complex and bulky setup.
- Strain gauges are widely used in various applications, such as measuring stress, force, pressure, torque, displacement, acceleration, vibration, etc. They are also used in experimental mechanics, structural analysis, material testing, biomedical engineering, etc. .

Force (Proving Ring) and Torques (Prony Brake Dynamometer)

- Force is the reaction between two bodies or components, which can be either tensile (pull) or compressive (push).
- Torque is the turning force applied to a rotating shaft, which is the product of force and distance from the axis of rotation.
- A proving ring is a device used to measure force or load by using a calibrated elastic ring that deforms under the applied force and indicates the magnitude of the force by a dial gauge or a micrometer.
- A prony brake is a device used to measure torque or power by using a frictional brake that clamps around a rotating shaft and creates a resistance that is proportional to the applied force and the coefficient of friction.
- The force applied by the prony brake is measured by a spring balance or a standard mass at the end of a lever arm, and the torque is calculated by multiplying the force by the lever arm length.
- The power transmitted by the shaft is calculated by multiplying the torque by the angular speed of the shaft.

Concepts of accuracy for the notes of the Unit 5 - Introduction to Measurement and Mechatronics in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

- Accuracy is the degree of closeness of a measurement to the true or accepted value of the quantity measured .
- Accuracy is affected by systematic errors, which are caused by factors such as calibration, environmental conditions, or instrument design .
- Accuracy can be expressed as a percentage of error, which is the ratio of the absolute error to the true value, multiplied by 100 .
- Accuracy can also be expressed as a tolerance, which is the range of values within which the true value is expected to lie .
- Accuracy can be improved by using more precise instruments, reducing systematic errors, and applying correction factors .
- Accuracy is different from precision, which is the degree of repeatability or consistency of a measurement .
- Precision is affected by random errors, which are caused by factors such as noise, fluctuations, or human errors .
- Precision can be expressed as a standard deviation, which is a measure of how much the measurements vary from the mean value .
- Precision can also be expressed as a resolution, which is the smallest change in the quantity measured that can be detected by the instrument .
- Precision can be improved by using more sensitive instruments, reducing random errors, and increasing the number of measurements .
- Accuracy and precision are both important for reliable and valid measurements, but they are not the same .
- A measurement can be accurate but not precise, precise but not accurate, both accurate and precise, or neither accurate nor precise .
- A measurement system should aim to achieve high accuracy and precision, but there may be trade-offs between them depending on the application and the resources available .

Precision and Resolution

- Precision is the amount of information that is conveyed in terms of digits. It refers to the resolution or limit of the measurement.
- Precision is also the degree of repeatability of a measurement, i.e., how reliably it can reproduce the same measurement over and over.
- Precision is independent of accuracy, which denotes how close the measured value is to the true value of a given quantity.
- Precision can be expressed by the number of significant figures, the standard deviation, or the coefficient of variation of a set of measurements.

- Resolution is the smallest increment an instrument can detect and display, such as hundredths, thousandths, or millionths.
- Resolution is also the total weighing range of a scale divided by the readability of the display.
- Resolution determines the smallest change in the quantity being measured that can be detected by the instrument.
- Resolution can be limited by the noise, the sensitivity, or the digitization of the instrument.
- Resolution can be improved by increasing the number of digits, the signal-to-noise ratio, or the sampling rate of the instrument.

Introduction to Mechatronic Systems

- A mechatronic system is a system that integrates mechanical, electrical, and computer engineering components to perform a desired function.
- Mechatronic systems can be found in various applications, such as robotics, automotive, aerospace, biomedical, manufacturing, and consumer products.
- Mechatronic systems are designed using a multidisciplinary approach that considers the interactions and trade-offs among different subsystems and components.
- Mechatronic systems typically consist of four main elements: sensors, actuators, controllers, and interfaces.
 - Sensors are devices that measure physical quantities, such as position, velocity, force, temperature, pressure, etc., and convert them into electrical signals.
 - Actuators are devices that convert electrical signals into physical actions, such as motion, force, torque, etc., to control the output of the system.
 - Controllers are devices that process the sensor signals and generate the actuator signals, using algorithms, logic, or feedback, to achieve the desired system performance and behavior.
 - Interfaces are devices that enable the communication and interaction between the system and the user, the environment, or other systems, using displays, keyboards, switches, wireless networks, etc.
- Mechatronic systems are characterized by some key features, such as:
 - Synergy: the system performance is greater than the sum of the individual component performances, due to the integration and coordination of different domains and disciplines.
 - Adaptability: the system can adjust its behavior and parameters according to the changes in the operating conditions, the user preferences, or the external disturbances, using feedback, learning, or optimization techniques.
 - Intelligence: the system can perform complex tasks, such as perception, decision making, planning, or learning, using artificial intelligence.

- gence, machine learning, or computer vision methods.
- Autonomy: the system can operate independently, without human intervention or supervision, using self-diagnosis, self-repair, or self-organization capabilities.

Evolution for the notes of the Unit 5 - Introduction to Measurement and Mechatronics in the subject of FUNDAMENTALS OF MECHANICAL ENGINEERING

- Measurement and mechatronics are two interrelated fields of engineering that involve the application of sensors, actuators, microprocessors, and control systems to various mechanical systems and processes.
- Measurement is the process of obtaining quantitative information about physical quantities, such as length, mass, force, temperature, etc. Measurement involves the use of instruments, standards, units, and methods to ensure accuracy, precision, reliability, and traceability of the data.
- Mechatronics is the integration of mechanical, electrical, and computer engineering to design, develop, and control smart and efficient systems, such as robots, cars, machine tools, cameras, etc. Mechatronics involves the use of sensors and measurement systems to monitor and feedback the system state, actuators and drive systems to manipulate and control the system output, and microprocessors and software to coordinate and optimize the system performance.
- The evolution of measurement and mechatronics can be traced back to the development of various technologies and disciplines over time, such as:
 - The invention of the first mechanical clocks in the 13th century, which enabled the measurement of time and the synchronization of events.
 - The emergence of the scientific method and the standardization of units and measurements in the 17th and 18th centuries, which facilitated the advancement of science and engineering.
 - The discovery of electricity and magnetism and the development of electrical and electronic devices and circuits in the 19th and 20th centuries, which enabled the measurement and control of electrical quantities and signals.
 - The invention of the transistor and the microprocessor and the development of digital and computer technology in the 20th and 21st centuries, which enabled the processing and communication of data and information and the automation and intelligence of systems and processes.
- The evolution of measurement and mechatronics has also been influenced by the challenges and demands of various domains and applications, such as:

- The industrial revolution and the mass production of goods and services, which required the measurement and control of quality, efficiency, and productivity.
- The aerospace and defense sectors and the exploration of space and warfare, which required the measurement and control of complex and dynamic systems and environments.
- The biomedical and health sectors and the improvement of human health and well-being, which required the measurement and control of physiological and biological parameters and functions.
- The consumer and entertainment sectors and the enhancement of human comfort and enjoyment, which required the measurement and control of user preferences and interactions.

Scope of Measurement and Mechatronics

- Measurement and mechatronics are two interrelated fields of engineering that deal with the design, development and application of systems that integrate mechanical, electrical, electronic and software components.
- Measurement is the process of obtaining, processing and interpreting data about physical quantities, such as temperature, pressure, displacement, force, etc. Measurement systems consist of sensors, transducers, signal conditioning circuits, data acquisition devices, and display or recording devices.
- Mechatronics is the synergistic combination of mechanical engineering, electronics, control engineering, and computer science, applied to the design of intelligent machines and systems. Mechatronics systems include robots, automated manufacturing systems, smart appliances, biomedical devices, etc.
- The scope of measurement and mechatronics is very broad and covers various domains of engineering and science, such as:
 - Industrial automation and control: Mechatronics systems are widely used to automate and optimize industrial processes, such as assembly, packaging, welding, machining, etc. Measurement systems are essential to monitor and control the quality, efficiency and safety of these processes.
 - Robotics and artificial intelligence: Mechatronics systems are the basis of robotics, which is the branch of engineering that deals with the design, construction and operation of machines that can perform tasks autonomously or semi-autonomously. Measurement systems provide the feedback and sensory information that enable robots to interact with their environment and perform tasks such as navigation, manipulation, recognition, etc.

- Automotive and aerospace engineering: Mechatronics systems are involved in the design and development of vehicles and aircrafts, such as cars, trucks, buses, trains, planes, helicopters, rockets, etc. Measurement systems are used to monitor and control the performance, reliability and safety of these vehicles and aircrafts, such as engine, transmission, suspension, braking, steering, etc.
- Biomedical engineering and health care: Mechatronics systems are applied to the design and development of medical devices and equipment, such as prosthetics, implants, surgical instruments, diagnostic tools, etc. Measurement systems are used to measure and analyze the physiological parameters and functions of the human body, such as blood pressure, heart rate, temperature, etc.
- Consumer electronics and entertainment: Mechatronics systems are integrated into various consumer electronic devices and products, such as mobile phones, computers, cameras, gaming consoles, etc. Measurement systems are used to provide the user interface and functionality of these devices and products, such as touch screen, motion sensor, camera, microphone, etc.
- The scope of measurement and mechatronics is constantly expanding and evolving, as new technologies, applications and challenges emerge in the fields of engineering and science. Mechatronics engineers and measurement specialists are in high demand in various industries and sectors, such as manufacturing, transportation, energy, defense, health care, education, etc.